



ACE-ProtoNet: Adaptive covariance eigen-gate and uncertainty-aware prototype learning for coronary artery segmentation

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ABSTRACT

Accurate segmentation of coronary arteries from Coronary CT Angiography (CCTA) is essential for quantitative stenosis evaluation, plaque characterization, and surgical planning. However, low vessel-to-background contrast, high anatomical variability, and the complex, tree-like vascular morphology pose significant challenges to automated segmentation. To address these challenges, we propose ACE-ProtoNet, a unified framework that couples an Adaptive Covariance Eigen-Gate (ACE-Gate) with an Uncertainty-aware Prototype Learning Head (UPL-Head) to achieve robust and accurate coronary artery segmentation. The architecture is built upon a parallel dual-encoder backbone which synergizes a partially frozen Vision Foundation Model (VFM) for global structural encoding with a trainable CNN for fine-grained local feature extraction. To reconcile the heterogeneity between these feature streams, the ACE-Gate explicitly models inter-channel dependencies through covariance analysis and eigenvalue decomposition, yielding statistically grounded, channel-wise gating for principled feature integration. Meanwhile, the UPL-Head leverages voxel-wise predictive uncertainty to modulate prototype-guided attention and dynamically update prototypes during training, thereby enhancing representation robustness in hard-to-classify regions and improving overall segmentation accuracy. Extensive experiments on two in-house and four public datasets demonstrate that ACE-ProtoNet consistently outperforms twelve state-of-the-art methods across multiple metrics, exhibiting superior cross-domain generalization as well as strong cross-modality and cross-anatomy transferability. The code is available at .

1. Introduction

Coronary Artery Disease (CAD) persists as a leading global cause of mortality, necessitating precise and timely diagnosis to guide effective clinical decision-making (Virani et al., 2021). Coronary CT Angiography (CCTA) has become the primary non-invasive modality for visualizing the coronary artery tree, enabling quantitative analyses such as stenosis assessment and surgical planning (Marano et al., 2020). Consequently, precise segmentation of coronary arteries is a critical prerequisite for computer-aided diagnosis. However, manual delineation is prohibitively time-consuming and prone to inter-observer variability, emphasizing the need for accurate automated methods.

Despite recent advances in deep learning (Moeskops et al., 2016; Song et al., 2022; Dong et al., 2023), accurate coronary artery segmentation remains challenging due to several intrinsic factors, as illustrated in Fig. 1: (1) low vessel-to-background contrast; (2) significant inter-patient anatomical variability; and (3) complex, tree-like vascular topology with blurred boundaries. These factors often lead to fragmented or incomplete segmentations, motivating the development of advanced learning-based methods.

Early models based on CNNs (Moeskops et al., 2016; Song et al., 2022; Dong et al., 2023), while effective at capturing local details, are known to be constrained by a limited effective receptive field (Luo et al., 2016), which restricts their ability to capture long-range spatial

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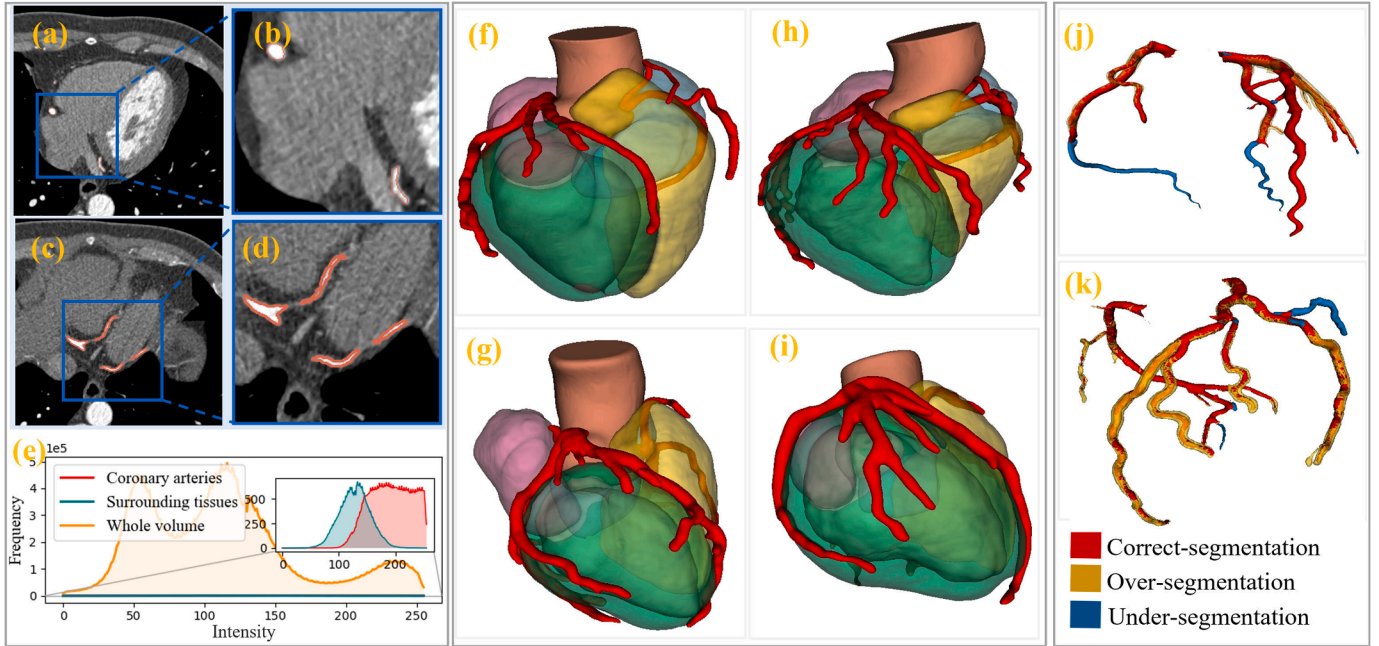


Fig. 1. Challenges in coronary artery segmentation from CCTA. (a, c) Representative axial slices from a CCTA scan, with (b, d) showing magnified views of the boxed regions to highlight the low contrast and ambiguous boundaries of coronary arteries (ground truth in red). (e) Intensity histograms show significant overlap between coronary arteries (red) and surrounding tissues (cyan), quantitatively demonstrating the low vessel-to-background contrast. (f-i) 3D renderings of four different patient cases, highlighting the significant inter-patient anatomical variability and complex vascular topology. (j, k) Examples of segmentation errors from a baseline method, showing correctly-segmented (red), over-segmented (orange), and under-segmented (blue) regions, which result in fragmentation and loss of continuity.

dependencies. Transformer-based architectures improve global context modeling but can compromise fine-grained spatial precision, partly due to the lack of inductive biases inherent in convolutions (Dosovitskiy et al., 2020; Chen et al., 2021). This has naturally led to the development of hybrid architectures designed to leverage the complementary strengths of both paradigms (Pan et al., 2022). Building upon the Transformer trend, Vision Foundation Models (VFM), such as the Segment Anything Model (SAM) (Kirillov et al., 2023), have demonstrated particularly strong generalization capabilities. The current state-of-the-art often involves coupling a powerful, pre-trained VFM with a task-specific CNN (Chen et al., 2024a). However, most existing methods (Pan et al., 2022; Wu et al., 2022) either operate on 2D slices with post-hoc merging or rely on implicit or heuristic strategies (e.g., naive concatenation), neglecting 3D inter-slice continuity and failing to reconcile heterogeneous feature statistics.

Another critical yet underexplored challenge lies in predictive uncertainty. In clinically ambiguous regions, such as heavily calcified areas, distal vessels, or low-contrast regions, even state-of-the-art models produce unreliable predictions. This motivates the need for uncertainty-aware mechanisms that can not only identify but also actively leverage uncertain predictions to improve robustness and reliability.

To address these challenges, we propose ACE-ProtoNet, a unified framework that couples an Adaptive Covariance Eigen-Gate (ACE-Gate) with an Uncertainty-aware Prototype Learning Head (UPL-Head). Its architecture is built upon a parallel dual-encoder backbone that integrates a 3D foundation model-driven encoder, fine-tuned from SAM-Med3D (Wang et al., 2023), with a 3D U-Net-based shape encoder. This design is further enhanced by two key innovations targeting the core challenges of heterogeneous feature fusion and predictive uncertainty. First, the ACE-Gate explicitly models inter-channel dependencies via covariance analysis and generates adaptive gating weights from eigenvalues, enabling stable and principled integration of heterogeneous feature streams. Second, the UPL-Head leverages voxel-wise predictive uncertainty to modulate prototype-guided attention and dynamically update prototypes, thereby enhancing robustness in hard-to-classify regions.

Overall, ACE-ProtoNet unifies the ACE-Gate and the UPL-Head within a hybrid end-to-end framework, providing a principled and uncertainty-aware solution for accurate and robust coronary artery segmentation.

The main contributions of this work are as follows:

- We propose a dual-encoder architecture that synergizes a partially frozen VFM for global structural encoding with a trainable CNN-based encoder for capturing fine-grained local features.
- We develop the ACE-Gate, a statistically principled fusion mechanism that models inter-channel dependencies via covariance and eigenvalue analysis, producing channel-wise gating weights for optimal feature integration.
- We design the UPL-Head, which leverages voxel-wise predictive uncertainty to modulate prototype-guided attention and guide online prototype updates, ensuring robust representation in ambiguous regions.
- Extensive experiments on two in-house and three public datasets demonstrate that ACE-ProtoNet consistently outperforms state-of-the-art methods and exhibits superior cross-domain generalization, as well as strong cross-modality and cross-anatomy transferability.

2. Related works

This section reviews the literature relevant to our proposed ACE-ProtoNet, focusing on three key areas: coronary artery segmentation, advanced feature fusion techniques, and the integration of prototype and uncertainty learning.

2.1. Coronary artery segmentation

Coronary artery segmentation from CCTA plays a pivotal role in quantitative cardiovascular assessment, yet it remains a challenging task. Before the deep learning era, classical approaches, most notably the foundational multi-scale vessel enhancement filters (Frangi et al., 1998), as well as model-based tracking, region growing, and level-set

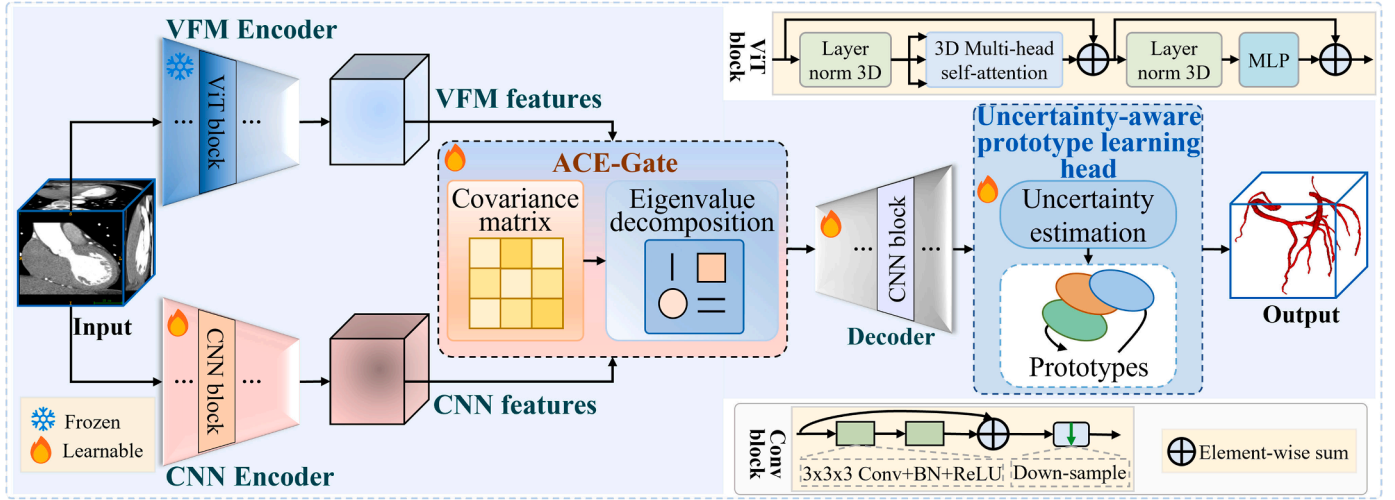


Fig. 2. Overall architecture of the proposed ACE-ProtoNet. The model processes an input 3D CTA volume through a parallel dual-encoder backbone. The upper branch employs a partially frozen VFM encoder that captures global structural representations, while the lower branch uses a fully trainable CNN encoder to extract fine-grained, task-specific local features. To reconcile this heterogeneity, the proposed ACE-Gate provides a principled fusion mechanism by modeling inter-channel dependencies via covariance analysis and eigenvalue decomposition. The fused features are then decoded and refined by the UPL-Head, which leverages predictive uncertainty and class-level prototypes to enhance segmentation accuracy.

methods (Manniesing and Niessen, 2004; Wang et al., 2012; Yang et al., 2012) were widely adopted for their efficiency and minimal data requirements. However, these techniques rely heavily on handcrafted features and manual initialization, making them highly sensitive to imaging noise, anatomical variability, and pathological changes such as stenosis and calcification.

The advent of deep learning has led to remarkable progress in volumetric vessel segmentation (Jin et al., 2025). CNNs, particularly 3D U-Net (Ronneberger et al., 2015) and its derivatives (Moeskops et al., 2016; Song et al., 2022; Dong et al., 2023), have become dominant due to their strong representation capacity. Extensions such as 3D-FFR-UNET (Song et al., 2022) employs dense convolutional blocks for multi-scale feature aggregation, while attention mechanisms further enhance fine anatomical details (Dong et al., 2023). To further improve vessel connectivity and alleviate fragmentation, recent works have also incorporated topology-aware tracking (Li et al., 2021) and geometry-based vectorization (Yang et al., 2024). Despite these advances, CNNs are known to be constrained by a limited effective receptive field (Luo et al., 2016), which restricts their ability to capture long-range dependencies that are crucial for preserving the continuity of complex vascular trees.

Transformer-based architectures, in contrast, excel at modeling global dependencies and structural priors. Representative models such as UNETR (Hatamizadeh et al., 2022), TransUNet (Chen et al., 2024b), and nnFormer (Zhou et al., 2023) have achieved competitive segmentation performance. Nevertheless, their self-attention mechanisms may under-represent local spatial granularity, partly due to the lack of inductive biases inherent in convolutions (Dosovitskiy et al., 2020; Chen et al., 2021), leading to suboptimal delineation of high-curvature vessel boundaries and small distal branches.

This trade-off between local precision and global coherence has motivated hybrid architectures that integrate the complementary strengths of CNNs and Transformers. For example, Pan et al. (2022) introduced a cross-transformer network that combines U-Net for local representation and Transformers for global context modeling. More recently, vision foundation models (VFMs) have reshaped the paradigm by leveraging large-scale pretraining. However, models such as SAM (Kirillov et al., 2023), trained on natural images, suffer from severe domain gaps when transferred to medical imaging. Hybrid variants like Ensembled-SAMs (Chen et al., 2024a) typically operate in a 2D slice-wise manner and rely on result-level merging, limiting their ability to model volumet-

ric consistency due to the absence of inter-slice feature encoding. The emergence of 3D VFMs, such as SAM-Med3D (Wang et al., 2023), effectively mitigates representation-level domain shift through large-scale medical pretraining and enables native 3D processing. Nevertheless, these models primarily provide generic semantic representations and are not inherently optimized for capturing fine-grained local anatomy or topology-sensitive structures. As a result, simply substituting the backbone does not automatically resolve the challenge of principled feature-level fusion. Integration of high-level representations from VFMs with CNN-based anatomical features often still relies on implicit or heuristic strategies (e.g., channel-wise concatenation in FAT-Net (Wu et al., 2022)), which fail to explicitly model or reconcile the heterogeneous feature statistics across branches.

Building on these advancements, we develop ACE-ProtoNet, a hybrid framework featuring a parallel dual-encoder backbone that integrates a 3D foundation model-driven encoder (fine-tuned from SAM-Med3D) with a 3D U-Net-based shape encoder. Its design is enhanced by two key innovations: the ACE-Gate, a statistically guided fusion mechanism for principled feature integration, and the UPL-Head, which explicitly models predictive uncertainty to enhance robustness, particularly in delineating the small, ambiguous distal vessel segments where previous methods often struggle.

2.2. Feature fusion strategies in hybrid models

Effective feature fusion is a critical component in hybrid architectures, particularly when integrating heterogeneous representations from VFMs and CNNs. Attention-based mechanisms have become the dominant paradigm, learning data-dependent relevance weights for feature streams. Channel attention, such as SENet (Hu et al., 2018) and ECA-Net (Wang et al., 2020), compresses spatial information into compact descriptors but risks losing fine-grained details. Query-driven attention, including Transformers (Vaswani et al., 2017) and TransUNet (Chen et al., 2021), enables richer contextual modeling yet incurs high computational costs and exhibits sensitivity to the quality of learned queries. Despite their successes, these methods depend solely on learned transformations to approximate feature relevance, lacking the explicit statistical grounding found in higher-order pooling strategies (Lin et al., 2015) and adaptive balancing between heterogeneous feature streams.

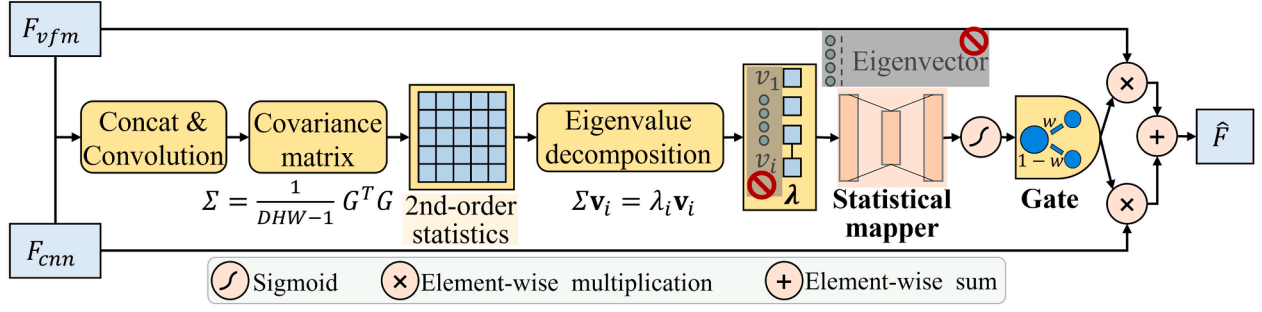


Fig. 3. Structure of the ACE-Gate. The mechanism operates in three statistically grounded stages: **(1) Common space projection and statistical modeling:** input features are projected into a unified space where the inter-channel covariance matrix is computed to capture second-order statistics (global variance structure). **(2) Eigenvalue-guided weight generation:** eigenvalue decomposition extracts the global variance profile (λ). A statistical mapper (MLP) then maps these eigenvalues into channel-wise importance weights (w), bridging the gap between global statistical structure and channel-wise fusion relevance. **(3) Channel-wise adaptive fusion:** these weights orchestrate a complementary gating mechanism (w and $1 - w$), dynamically balancing VFM-derived global semantics and CNN-based local details to produce the final fused output $\hat{F}$.

Second-order statistics offer a principled alternative for feature integration. Techniques such as covariance pooling and bilinear transformations (Lin et al., 2015; Gao et al., 2019) capture inter-channel dependencies more effectively, but their direct application to segmentation is computationally expensive and, more importantly, lacks explicit mechanisms to reconcile global structural features with local spatial details.

To address these limitations, we propose the ACE-Gate module, a statistically grounded fusion strategy. ACE-Gate models inter-channel dependencies via covariance estimation and translates the resulting eigenvalues into channel-wise gating weights, enabling adaptive stream balancing. This design achieves a soft yet principled selection between global and local features, leading to more stable and robust integration.

2.3. Prototype and uncertainty-aware learning

Prototype learning enhances feature discriminability in medical imaging by summarizing class-level information into representative exemplars (prototypes) for classification in a learned embedding space. While early works used prototypes mainly as auxiliary regularizers (Snell et al., 2017; Zhang et al., 2020), recent approaches embed them directly into the prediction pathway to sharpen semantic boundaries and improve small-structure segmentation (Zhou et al., 2024b; Yu et al., 2025). Despite these advances, conventional prototypes often struggle in ambiguous regions, where they fail to remain representative of challenging samples.

Uncertainty modeling offers a complementary mechanism to address these ambiguities. A wide range of techniques have been employed to highlight unreliable regions, guide network optimization, and calibrate model predictions in medical imaging. These include Bayesian approximation (Kwon et al., 2020; Nair et al., 2020), entropy- and variance-based estimation (Dong et al., 2024; Mehrtash et al., 2020), and Dirichlet-based evidential learning frameworks (Xu et al., 2024; Dong et al., 2025). While effective, these two paradigms, prototype learning and uncertainty modeling, have largely been explored in isolation, leaving their synergistic potential underexploited.

Our work bridges this gap with the proposed UPL-Head. In our framework, voxel-wise predictive uncertainty is treated as an indicator of structural ambiguity (i.e., sample hardness). In complex vascular structures, high-uncertainty regions predominantly correspond to thin distal branches or vessel boundaries that are under-represented by standard prototype learning schemes. By explicitly incorporating uncertainty into both prototype updating and prototype-guided prediction, UPL-Head enables prototypes to remain robust and representative, particularly for complex vascular topologies.

3. Proposed method

3.1. Network architecture

We propose ACE-ProtoNet, a hybrid end-to-end framework for accurate and robust coronary artery segmentation, as illustrated in Fig. 2. The architecture comprises four key components: a dual-encoder backbone, the ACE-Gate module, a decoder, and a UPL-Head. The dual-encoder backbone synergizes a global encoder, implemented as a partially frozen SAM-Med3D model (Wang et al., 2023) (fine-tuning only the final two ViT blocks), with a trainable 3D U-Net-style CNN encoder for fine-grained spatial details. The heterogeneous features are fused by ACE-Gate, a statistically principled module that computes inter-channel covariance and performs Eigenvalue Decomposition (EVD) to derive variance indicators for an adaptive gating mechanism. The fused features are progressively upsampled by a U-Net-style decoder with skip connections from the CNN encoder to restore high-resolution details. Finally, UPL-Head refines the prediction by incorporating voxel-wise uncertainty into both prototype-guided attention and online prototype updating, suppressing unreliable responses and emphasizing challenging regions. This cohesive architecture enables ACE-ProtoNet to jointly optimize feature fusion and uncertainty modeling, yielding superior generalization and clinical reliability.

3.2. Adaptive covariance eigen-gate module

Hybrid architectures face a fundamental challenge in fusing heterogeneous features: VFMs provide global structural features, whereas CNNs capture fine-grained spatial details, typically with distinct statistical distributions. Naïve fusion strategies (e.g., concatenation or addition) may lead to feature dominance and degraded discriminability, while attention-based fusion without statistical grounding often yields unstable optimization.

To address these limitations, we propose the ACE-Gate, a statistically principled fusion mechanism that reconciles heterogeneous representations by explicitly modeling inter-channel dependencies. ACE-Gate adaptively integrates encoder features through covariance estimation and eigenvalue-guided gating, and operates in three key stages, as illustrated in Fig. 3.

3.2.1. Common space projection and statistical modeling

This stage transforms the dual-encoder features into a unified statistical representation for adaptive fusion. Let $F_{vfm}, F_{cnn} \in \mathbb{R}^{D \times H \times W \times C_{ch}}$ denote the feature maps from the VFM and CNN encoders, respectively, where D , H , and W denote the spatial dimensions (depth, height, and width), and C_{ch} is the number of feature channels. Here, F_{vfm} represents the VFM feature map that has been projected via a $1 \times 1 \times 1$

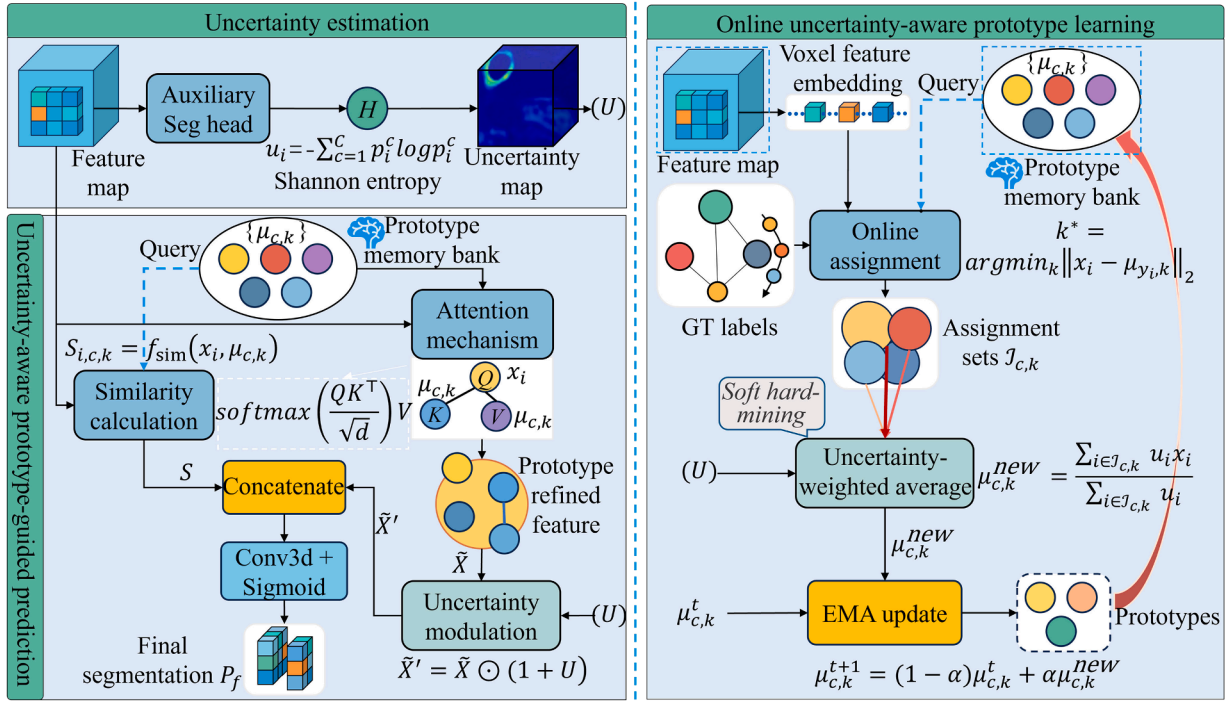


Fig. 4. Structure of the UPL-Head. (1) **Uncertainty estimation (Top-Left):** An auxiliary head computes a voxel-wise uncertainty map (U) via Shannon entropy to identify ambiguous regions. (2) **Uncertainty-aware prototype-guided prediction (Bottom-Left):** Decoder features are refined via attention, where voxels act as queries and prototypes as keys/values. The resulting features ($\tilde{X}$) are modulated by uncertainty ($\tilde{X}' = \tilde{X} \odot (1 + U)$) to amplify global priors. In parallel, an explicit similarity map (S) is computed. These are concatenated and passed through a 3D convolution with sigmoid to produce the final segmentation (P_f). (3) **Online uncertainty-aware prototype learning (Right, Training Only):** Voxels are assigned to the nearest prototype (k^*) and aggregated via uncertainty-weighted averaging (performing *soft hard-mining*). This effectively prioritizes “hard samples” (high uncertainty) to enhance prototype discriminability, followed by an EMA update. During inference, the prototype bank is fixed.

convolution to match the dimensionality of F_{cnn} . The two feature tensors are then concatenated along the channel dimension and passed through a shared $1 \times 1 \times 1$ convolution to obtain a unified representation $X \in \mathbb{R}^{D \times H \times W \times C_{ch}}$. This step aligns the semantic spaces and facilitates joint statistical analysis.

Next, X is reshaped into $X' \in \mathbb{R}^{DHW \times C_{ch}}$ and is subsequently mean-centered by subtracting the column-wise mean $\bar{X}'$ to obtain $G = X' - \bar{X}'$. The inter-channel covariance matrix $\Sigma \in \mathbb{R}^{C_{ch} \times C_{ch}}$ is computed as:

$$\Sigma = \frac{1}{DHW - 1} G^T G \quad (1)$$

which captures the second-order statistical dependencies and co-activation patterns among feature channels. Such covariance statistics provide a robust basis for adaptive gating.

3.2.2. Eigenvalue-guided weight generation

With the covariance matrix Σ established, this stage translates second-order statistics into a channel-wise gating vector. We perform EVD on Σ to obtain eigenvalues $\{\lambda_i\}$ and corresponding eigenvectors $\{v_i\}$ satisfying $\Sigma v_i = \lambda_i v_i$. While the eigenvectors characterize principal component directions, they are rotation-variant and sensitive to feature perturbations. Therefore, we exclusively utilize the eigenvalues, which summarize the variance distribution along principal components (Abdi and Williams, 2010) and serve as compact, rotation-invariant descriptors of global feature variance.

This design fundamentally distinguishes ACE-Gate from conventional attention mechanisms (e.g., SE-Net (Hu et al., 2018)), which rely on global average pooling to capture first-order statistics (mean activations). By contrast, eigenvalues encode second-order statistics that characterize how information is distributed and correlated across channels, independent of absolute activation magnitude and spatial arrangement.

Since eigenvalues represent variance along principal components rather than explicit semantic channels, they do not have a one-to-one

correspondence with the original feature channels. To bridge this representational gap, we employ a lightweight MLP as a *statistical mapper*. Importantly, this MLP is constrained to operate exclusively on the full set of eigenvalues, without access to raw spatial features or channel-wise activations. Its role is to learn a non-linear mapping from the global variance structure to channel-wise relevance weights:

$$w = \sigma(\text{MLP}(\lambda)) \in [0, 1]^{C_{ch} \times 1} \quad (2)$$

where the Sigmoid function bounds the weights within $[0, 1]$ to enable soft gating. Crucially, although this mapping is learnable, its effective degrees of freedom are fundamentally limited by the low-dimensional statistical input. This constrains the module to guide feature fusion based on second-order statistical structure rather than content-driven attention, enforcing the intended inductive bias.

3.2.3. Channel-wise adaptive fusion

The generated gating vector w orchestrates the final fusion of the dual-encoder features. The fused representation $\hat{F}$ is obtained as:

$$\hat{F} = (F_{\text{vfm}} \odot w) + (F_{\text{cnn}} \odot (1 - w)) \quad (3)$$

where $\odot$ denotes element-wise multiplication with spatial broadcasting. This mechanism acts as a soft selection gate: when $w_i \rightarrow 1$, the fusion emphasizes VFM-derived global semantics; when $w_i \rightarrow 0$, it favors CNN-based local details. By grounding feature fusion in covariance-driven statistics, ACE-Gate ensures stable, principled, and adaptive integration across heterogeneous feature streams.

3.3. Uncertainty-aware prototype learning head

To further enhance segmentation accuracy in ambiguous vascular regions, we design the UPL-Head, a unified module that seamlessly integrates prototype learning with uncertainty modeling. As shown in Fig. 4,

this head comprises three interconnected components: (1) uncertainty estimation, (2) uncertainty-aware prototype-guided prediction, and (3) online uncertainty-aware prototype learning.

3.3.1. Uncertainty estimation

Reliable uncertainty quantification is essential for guiding predictions in ambiguous regions. To decouple uncertainty estimation from the final segmentation output, we attach an auxiliary segmentation head to the last decoder layer. This head produces a voxel-wise categorical distribution $\mathbf{p}_i = \{p_i^{(c)}\}_{c=1}^C$ for each voxel i .

Voxel-wise uncertainty can be estimated using various techniques, such as entropy-based measures or Dirichlet-based evidential learning frameworks. In this work, we compute voxel-wise uncertainty using Shannon entropy for its simplicity and efficiency; this component is interchangeable with other uncertainty metrics:

$$u_i = - \sum_{c=1}^C p_i^{(c)} \log(p_i^{(c)}) \quad (4)$$

where u_i denotes the predictive uncertainty of voxel i , and C is the number of classes. The resulting uncertainty map, $\mathbf{U} = u_i \mid i \in \Omega \in \mathbb{R}^{D \times H \times W}$, is normalized to $[0, 1]$ by dividing by its theoretical maximum, $\log(C)$. This spatial map highlights regions of low model confidence and can subsequently guide downstream processes, such as prototype adaptation or adaptive feature weighting.

3.3.2. Uncertainty-aware prototype-guided prediction

This component enhances prediction by synergizing two complementary prototype-based cues: (1) an explicit voxel-to-prototype similarity map, and (2) a context-aware feature representation refined by uncertainty-modulated attention. Given the decoder feature map $\mathbf{X} \in \mathbb{R}^{D \times H \times W \times C_{ch}}$, we first compute a similarity map $\mathbf{S}$ where each element $s_{i,c,k}$ measures the cosine similarity between a voxel feature $\mathbf{x}_i$ and a prototype $\mu_{c,k}$ from a dynamic memory bank $\{\mu_{c,k}\}_{c=1,\dots,C;k=1,\dots,K}$, where K is the number of prototypes per class:

$$s_{i,c,k} = \frac{\mathbf{x}_i \cdot \mu_{c,k}}{\|\mathbf{x}_i\| \|\mu_{c,k}\|} \quad (5)$$

This yields a comprehensive map $\mathbf{S} \in \mathbb{R}^{D \times H \times W \times (C \cdot K)}$ that encodes explicit voxel-to-prototype affinities.

Concurrently, a prototype-based attention mechanism is employed, where voxel features $\mathbf{x}_i$ act as queries ($\mathbf{Q}$), and the prototypes $\{\mu_{c,k}\}$ serve as both keys ($\mathbf{K}$) and values ($\mathbf{V}$). The mechanism computes an attention-weighted aggregation of prototypes to form a context vector, which is then fused with the original voxel feature via a residual connection to produce the prototype-refined feature map $\tilde{\mathbf{X}}$. To explicitly incorporate uncertainty, we perform adaptive modulation:

$$\tilde{\mathbf{X}}' = \tilde{\mathbf{X}} \odot (1 + \mathbf{U}) \quad (6)$$

where $\odot$ is element-wise multiplication. Here, the uncertainty map functions as an adaptive attention gate. In regions with high structural ambiguity, this modulation explicitly upweights the prototype-refined features, thereby compelling the network to rely more heavily on global class-level priors rather than potentially unreliable local textures. This ensures that segmentation decisions in hard-to-classify areas are stabilized by the robust guidance of representative prototypes.

Finally, the uncertainty-modulated features $\tilde{\mathbf{X}}'$ and the explicit similarity map $\mathbf{S}$ are concatenated. This combined representation, which leverages both implicit contextual guidance and explicit similarity scores, is fed into a final 3D convolutional layer with a Sigmoid activation to produce the segmentation probability map $\mathbf{P}_f = \sigma(\text{Conv3d}(\text{Concat}(\tilde{\mathbf{X}}', \mathbf{S})))$.

3.3.3. Online uncertainty-aware prototype learning

To maintain representative prototypes throughout training, we employ an online uncertainty-guided update strategy that adaptively refines prototype representations. At each training iteration, voxel features $\mathbf{x}_i$ are assigned to the nearest prototype within their ground-truth

Algorithm 1 Uncertainty-aware prototype learning head.

Input: Decoder features $\mathbf{X}$; ground truth labels $\mathbf{Y}$; current prototypes $\{\mu_{c,k}\}$; momentum α

Output: Final prediction $\mathbf{P}_f$; updated prototypes $\{\mu_{c,k}\}$

1: **Uncertainty estimation**
2: $\mathbf{P}_{\text{aux}} \leftarrow \text{AuxiliarySegmentationHead}(\mathbf{X})$
3: **for** each voxel $i \in \mathcal{V}$ **do**
4: $u_i \leftarrow - \sum_c P_{\text{aux}}^{(i,c)} \log P_{\text{aux}}^{(i,c)}$
5: **end for**
6: Assemble uncertainty map $\mathbf{U} = \{u_i\}$ and normalize to $[0, 1]$
7: **Uncertainty-aware prototype-guided prediction**
8: $\mathbf{S} \leftarrow \text{ComputeSimilarityMap}(\mathbf{X}, \{\mu_{c,k}\})$
9: $\tilde{\mathbf{X}} \leftarrow \text{Attention}(\mathbf{X}, \{\mu_{c,k}\})$
10: $\tilde{\mathbf{X}}' \leftarrow \tilde{\mathbf{X}} \odot (1 + \mathbf{U})$ ▷ Uncertainty-guided modulation
11: $\mathbf{Z} \leftarrow \text{Concat}(\tilde{\mathbf{X}}', \mathbf{S})$
12: $\mathbf{P}_f \leftarrow \text{Sigmoid}(\text{Conv3D}(\mathbf{Z}))$
13: **Online prototype update (training only)**
14: Initialize empty assignment sets $\mathcal{I}_{c,k}$ for all c, k
15: **for** each voxel $i \in \mathcal{V}$ **do**
16: $k^* \leftarrow \arg\min_k \|\mathbf{x}_i - \mu_{y_i,k}\|_2$ ▷ Find nearest prototype in ground-truth class
17: Add $\mathbf{x}_i$ to $\mathcal{I}_{y_i,k^*}$
18: **end for**
19: **for** each class c and prototype k **do**
20: **if** $\mathcal{I}_{c,k} \neq \emptyset$ **then**
21: $\mu_{c,k}^{\text{new}} \leftarrow \frac{\sum_{\mathbf{x}_i \in \mathcal{I}_{c,k}} u_i \cdot \mathbf{x}_i}{\sum_{\mathbf{x}_i \in \mathcal{I}_{c,k}} u_i}$ ▷ Uncertainty-weighted centroid
22: $\mu_{c,k} \leftarrow (1 - \alpha)\mu_{c,k} + \alpha\mu_{c,k}^{\text{new}}$ ▷ EMA update
23: **end if**
24: **end for**

class. This process identifies the prototype index k^* that minimizes the Euclidean distance in the feature space:

$$k^* = \arg\min_k \|\mathbf{x}_i - \mu_{y_i,k}\|_2 \quad (7)$$

This assignment step forms the sets $\mathcal{I}_{c,k}$, which group features based on their proximity to each class prototype. A candidate prototype is then computed as an uncertainty-weighted average of its assigned features:

$$\mu_{c,k}^{\text{new}} = \frac{\sum_{i \in \mathcal{I}_{c,k}} u_i \mathbf{x}_i}{\sum_{i \in \mathcal{I}_{c,k}} u_i} \quad (8)$$

In coronary artery segmentation, voxels with high predictive uncertainty typically correspond to structurally ambiguous regions, such as thin distal vessels or boundary interfaces (“hard samples”). Standard average pooling tends to bias prototype updates toward abundant, easy-to-classify voxels, causing prototypes to lose representativeness for these fine-grained structures. By weighting feature aggregation proportionally to voxel-wise uncertainty u_i , our method explicitly performs *soft hard-example mining*. This strategy counteracts the dominance of frequent, easy-to-classify samples and encourages updated prototypes to shift toward the feature distribution of challenging vascular patterns, thereby enhancing their discriminative power in ambiguous regions.

Prototypes are then updated via an Exponential Moving Average (EMA):

$$\mu_{c,k} \leftarrow (1 - \alpha)\mu_{c,k} + \alpha\mu_{c,k}^{\text{new}} \quad (9)$$

where α is a momentum hyperparameter (typically set to 0.1). This EMA mechanism enables smooth adaptation to evolving features while avoiding drastic fluctuations, leading to more stable training. During inference, the prototype memory bank is frozen and the online update module is deactivated. During inference, the final segmentation relies solely on the uncertainty estimation and prototype-guided prediction components.

Algorithm 1 summarizes UPL-Head, which incorporates voxel-wise uncertainty into prototype-guided prediction and online prototype

Table 1
Summary of datasets used in this study.

Dataset	Organ	Modality	Cases	In-plane resolution (mm)	Z-axis resolution (mm)	Volume Size (voxels)	Public
CTA119	Coronary arteries	CCTA	119	0.30-0.43	0.3-0.6	512 × 512 × (210 – 380)	×
ASOCA	Coronary arteries	CCTA	40	0.30-0.40	0.625	512 × 512 × (168 – 334)	✓
ImageCAS	Coronary arteries	CCTA	1000	0.29-0.43	0.25-0.45	512 × 512 × (206 – 275)	✓
PVS203	Portal venous system	CT	203	0.52-0.73	0.6-2.0	512 × 512 × (210 – 444)	×
MIDAS	Cerebrovascular system	MRA	42	0.5	0.8	448 × 448 × 128	✓

learning, thereby achieving refined segmentation and stable prototype evolution.

3.4. Loss function

The ACE-ProtoNet framework is trained end-to-end using a composite training objective that jointly optimizes for voxel-wise segmentation accuracy and a discriminative, prototype-guided feature embedding space. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \beta \mathcal{L}_{\text{ppc}} \quad (10)$$

where β is a hyperparameter that balances the contribution of the two objectives.

Segmentation loss Following best practices for medical segmentation, the segmentation loss $\mathcal{L}_{\text{seg}}$ is a hybrid of two complementary functions (Mou et al., 2021). A Weighted Cross-Entropy (WCE) loss, $\mathcal{L}_{\text{WCE}}$, mitigates the class imbalance between vessel and background voxels, while a Dice Similarity Coefficient (DSC) loss, $\mathcal{L}_{\text{DSC}}$, maximizes the spatial overlap between predictions and ground truth to preserve vessel continuity and topology. The combined loss is:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{WCE}} + \mathcal{L}_{\text{DSC}} \quad (11)$$

Voxel-prototype contrastive loss To complement the pixel-level supervision of $\mathcal{L}_{\text{seg}}$, we introduce the voxel-prototype contrastive loss ($\mathcal{L}_{\text{ppc}}$) to explicitly structure the embedding space. For a given voxel feature $\mathbf{x}_i$, its positive prototype μ_{c^+,k^+} is the nearest prototype from its corresponding ground-truth class, identified during the online assignment step (Section 3.3.3). This deep metric learning loss then maximizes the posterior probability of $\mathbf{x}_i$ being assigned to its positive prototype, formulated via a softmax function over the entire prototype memory bank $\mathcal{M}$:

$$P(\mu_{c^+,k^+} | \mathbf{x}_i) = \frac{\exp(\text{sim}(\mathbf{x}_i, \mu_{c^+,k^+})/\tau)}{\sum_{\mu \in \mathcal{M}} \exp(\text{sim}(\mathbf{x}_i, \mu)/\tau)} \quad (12)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and τ is a temperature hyperparameter. The loss is the negative log-likelihood, averaged over all labeled voxels, $\mathcal{V}_{\text{label}}$:

$$\mathcal{L}_{\text{ppc}} = -\frac{1}{|\mathcal{V}_{\text{label}}|} \sum_{i \in \mathcal{V}_{\text{label}}} \log P(\mu_{c^+,k^+} | \mathbf{x}_i) \quad (13)$$

This objective pulls each embedding towards its positive prototype while pushing it away from all other negative prototypes, enforcing high intra-class compactness and inter-class separability in the latent space. The balancing weight β is empirically set to 0.2.

4. Experiment configurations

4.1. Datasets

We evaluate the proposed ACE-ProtoNet on six datasets spanning coronary, portal, cerebrovascular, and pulmonary airway regions, as summarized in Table 1. These include two in-house datasets (CTA119 and PVS203) and four publicly available benchmarks (ASOCA, ImageCAS, MIDAS, and ATM22). While the coronary artery datasets (CTA119, ASOCA, ImageCAS) are used to assess segmentation accuracy, the

PVS203, MIDAS, and ATM22 datasets enable evaluation of cross-organ transferability and broader applicability to general tubular structures.

CTA119 is a retrospective in-house dataset comprising 119 CCTA volumes acquired from a leading tertiary teaching hospital under standard clinical protocols. Ground-truth annotations of coronary arteries were created in a two-stage process: initial labeling by three experienced radiologists followed by consensus-based refinement to ensure annotation consistency and anatomical accuracy.

ASOCA is a public dataset released by the MICCAI 2020 automated segmentation of coronary arteries challenge (Gharleghi et al., 2023)¹. It contains 40 CCTA scans, half of which are from patients with CAD, including both normal and diseased cases, each with voxel-wise annotations generated by clinical experts.

ImageCAS is a large-scale benchmark dataset comprising 1000 CCTA volumes from patients diagnosed with coronary artery disease (Zeng et al., 2023)². Many subjects underwent early revascularization within 90 days after diagnosis, providing a diverse spectrum of coronary anatomies and lesion severities.

PVS203 is an in-house dataset composed of 203 abdominal contrast-enhanced CT scans, including 125 patients diagnosed with Portal Hypertension (PH) and 78 health checks. Manual annotations of the Portal Venous System (PVS), comprising the portal vein, splenic vein, and superior mesenteric vein, were created for each scan. Since vessel morphology in the PVS plays a key role in evaluating PH, this dataset serves as an essential benchmark for studying vascular alterations related to portal hypertension.

MIDAS is a publicly available dataset of 42 MRA volumes with manually labeled cerebrovascular structures³.

ATM22 comprises 299 thoracic CT scans from the ATM22 Challenge (Zhang et al., 2023)⁴. It provides voxel-wise annotations of the pulmonary airway tree, including the trachea and hierarchical bronchial branches, serving as a benchmark for segmenting non-vascular tubular structures with complex topologies.

4.2. Evaluation metrics

To comprehensively evaluate the performance of ACE-ProtoNet, we adopt a set of complementary evaluation metrics encompassing voxel-wise accuracy, boundary precision, and topological consistency.

Voxel-wise metrics We employ the DSC and Sensitivity (SE, also known as Recall or True Positive Rate (TPR)) to assess voxel-level segmentation accuracy. They are defined as:

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|}, \quad \text{SE} = \frac{|P \cap G|}{|G|} \quad (14)$$

where P denotes the predicted segmentation mask and G represents the ground truth. DSC evaluates the volumetric overlap between prediction and reference, while SE measures the proportion of true vessel voxels correctly identified by the model.

Boundary-based metric To quantify the precision of boundary localization, we compute the Average Symmetric Surface Distance (ASSD).

¹ <https://asoca.grand-challenge.org>

² <https://www.kaggle.com/datasets/xiaoweixumedicallai/imagecas>

³ <https://public.kitware.com/Wiki/TubeTK/Data>

⁴ <https://atm22.grand-challenge.org>

Table 2
Quantitative comparison of our method and eleven SOTA approaches on three benchmark datasets (CTA119, ASOCA, and ImageCAS). Best results are highlighted in **bold**.

Method	CTA119						ASOCA						ImageCAS					
	DSC (%)↑	SE (%)↓	ASDD (mm)↓	cDice (%)↑	SR (%)↑	SP (%)↑	DSC (%)↑	SE (%)↓	ASDD (mm)↓	cDice (%)↑	SR (%)↑	SP (%)↑	DSC (%)↑	SE (%)↓	ASDD (mm)↓	cDice (%)↑	SR (%)↑	SP (%)↑
UNet3D (2016)	81.69 ±0.72	87.18 ±0.75	1.61 ±0.12	88.72 ±0.68	87.79 ±0.75	89.67 ±0.81	84.81 ±0.84	87.42 ±0.81	1.40 ±0.13	87.92 ±0.72	87.12 ±0.80	88.73 ±0.78	82.18 ±0.77	85.71 ±0.72	1.63 ±0.15	86.24 ±0.63	85.27 ±0.70	87.23 ±0.68
CPNet (2020)	83.55 ±0.63	88.83 ±0.66	1.51 ±0.11	90.13 ±0.61	89.21 ±0.68	91.07 ±0.65	85.95 ±0.80	88.78 ±0.69	1.29 ±0.12	89.85 ±0.62	89.14 ±0.70	90.57 ±0.66	84.04 ±0.63	87.08 ±0.59	1.46 ±0.12	87.83 ±0.59	86.93 ±0.65	88.75 ±0.62
S ³ CA-Net (2024)	84.05 ±0.58	89.14 ±0.57	1.13 ±0.10	90.55 ±0.57	89.66 ±0.62	91.46 ±0.60	87.15 ±0.79	89.31 ±0.63	1.15 ±0.12	90.23 ±0.58	89.43 ±0.65	91.04 ±0.62	84.39 ±0.61	87.33 ±0.58	1.18 ±0.13	88.26 ±0.56	87.42 ±0.60	89.12 ±0.58
MS-DUNet (2025)	84.58 ±0.54	89.52 ±0.53	0.96 ±0.09	91.07 ±0.55	90.14 ±0.60	92.02 ±0.58	87.75 ±0.80	90.25 ±0.56	0.82 ±0.11	91.13 ±0.57	90.32 ±0.62	91.95 ±0.60	85.01 ±0.58	88.64 ±0.53	1.05 ±0.11	88.87 ±0.55	88.03 ±0.58	89.73 ±0.56
MedT (2021)	83.92 ±0.64	88.88 ±0.60	1.24 ±0.10	90.24 ±0.59	89.29 ±0.65	91.21 ±0.62	87.12 ±0.80	89.93 ±0.65	1.25 ±0.10	88.95 ±0.60	88.23 ±0.68	89.68 ±0.65	84.23 ±0.64	87.85 ±0.61	1.20 ±0.12	87.94 ±0.58	87.14 ±0.62	88.76 ±0.60
UNETR (2022)	83.25 ±0.57	88.91 ±0.56	1.35 ±0.11	90.39 ±0.58	89.46 ±0.62	91.34 ±0.60	86.32 ±0.82	88.64 ±0.61	1.23 ±0.12	89.87 ±0.59	89.17 ±0.65	90.58 ±0.62	84.62 ±0.60	88.06 ±0.57	1.18 ±0.12	88.27 ±0.56	87.46 ±0.60	89.09 ±0.58
TransUNet (2024)	84.56 ±0.62	89.73 ±0.53	1.21 ±0.09	90.87 ±0.57	89.94 ±0.60	91.82 ±0.58	87.40 ±0.81	89.19 ±0.57	1.17 ±0.10	89.92 ±0.58	89.24 ±0.62	90.61 ±0.60	84.74 ±0.55	88.13 ±0.53	1.15 ±0.11	88.82 ±0.56	88.01 ±0.60	89.64 ±0.58
nn-Former (2023)	84.63 ±0.56	89.54 ±0.50	1.07 ±0.11	91.13 ±0.54	90.21 ±0.58	92.07 ±0.55	87.52 ±0.76	90.26 ±0.53	0.97 ±0.11	90.78 ±0.56	90.13 ±0.60	91.44 ±0.58	84.86 ±0.52	88.32 ±0.50	1.08 ±0.11	89.08 ±0.57	88.27 ±0.60	89.91 ±0.58
Ensembled-SAMs(2024)	84.92 ±0.55	90.11 ±0.47	0.83 ±0.10	91.29 ±0.52	90.34 ±0.55	92.26 ±0.52	88.02 ±0.78	90.84 ±0.49	0.80 ±0.11	91.62 ±0.54	90.82 ±0.58	92.43 ±0.55	85.08 ±0.49	88.50 ±0.47	1.02 ±0.11	89.15 ±0.56	88.32 ±0.58	89.99 ±0.55
CS ² Net (2021)	84.36 ±0.61	90.13 ±0.45	1.01 ±0.10	91.31 ±0.55	90.38 ±0.58	92.26 ±0.56	87.05 ±0.82	90.05 ±0.47	0.91 ±0.12	90.49 ±0.56	89.71 ±0.60	91.28 ±0.58	84.62 ±0.47	87.82 ±0.45	1.05 ±0.11	87.93 ±0.57	87.13 ±0.60	88.74 ±0.58
3D-FFR-UNet (2022)	84.60 ±0.55	89.62 ±0.42	0.90 ±0.09	91.42 ±0.53	90.49 ±0.56	92.37 ±0.54	87.10 ±0.64	90.78 ±0.44	0.83 ±0.09	91.22 ±0.54	90.41 ±0.58	92.04 ±0.56	84.77 ±0.44	88.42 ±0.42	0.97 ±0.10	88.47 ±0.56	87.68 ±0.58	89.27 ±0.55
VSNet (2025)	85.10 ±0.51	90.61 ±0.39	0.76 ±0.08	91.65 ±0.52	90.71 ±0.55	92.61 ±0.52	88.15 ±0.72	91.02 ±0.42	0.74 ±0.10	91.69 ±0.54	90.93 ±0.55	92.46 ±0.55	85.12 ±0.41	88.70 ±0.41	0.95 ±0.10	88.92 ±0.55	88.13 ±0.55	89.73 ±0.55
Ours	87.34 ±0.43	92.72 ±0.36	0.71 ±0.08	92.79 ±0.51	91.91 ±0.55	93.69 ±0.50	90.18 ±0.57	93.54 ±0.42	0.65 ±0.09	93.62 ±0.53	92.84 ±0.56	94.41 ±0.52	86.73 ±0.34	90.53 ±0.41	0.89 ±0.10	90.83 ±0.54	90.04 ±0.55	91.63 ±0.54

P-values between Ours and each comparison method: <0.01 (DSC), <0.01 (SE), <0.01 (ASDD), <0.01 (cDice).

This metric reflects the average bidirectional distance between the surfaces of predicted and ground truth masks:

$$\text{ASSD} = \frac{1}{|P_S| + |G_S|} \left(\sum_{p \in P_S} \min_{g \in G_S} d(p, g) + \sum_{g \in G_S} \min_{p \in P_S} d(g, p) \right) \quad (15)$$

Here, P_S and G_S denote the set of surface voxels of the predicted and ground truth masks, respectively, and $d(\cdot, \cdot)$ is the Euclidean distance.

Topology-aware metric To evaluate how well vascular topology is preserved, particularly critical for tree-like structures such as coronary arteries, we adopt the centerline Dice (clDice) metric (Shit et al., 2021). Given the predicted segmentation V_P and ground truth V_L , the clDice score is computed as:

$$\text{clDice}(V_P, V_L) = 2 \cdot \frac{T_{\text{prec}}(S_P, V_L) \cdot T_{\text{sens}}(S_L, V_P)}{T_{\text{prec}}(S_P, V_L) + T_{\text{sens}}(S_L, V_P)} \quad (16)$$

where S_P and S_L represent the skeletonized centerlines of the prediction and reference, respectively. T_{prec} and T_{sens} denote the topological precision and sensitivity. clDice is particularly sensitive to fragmentation and discontinuities, making it a valuable complement to volumetric metrics.

To explicitly quantify vessel connectivity and fragmentation, as suggested in recent vessel segmentation studies (Zhao et al., 2022), we further decompose clDice into two skeleton-based connectivity metrics: Skeleton Recall (SR) and Skeleton Precision (SP), which correspond directly to the topological sensitivity (T_{sens}) and topological precision (T_{prec}) terms used in clDice. Specifically, SR measures the fraction of the ground-truth centerline covered by the prediction, whereas SP evaluates the proportion of the predicted skeleton that correctly lies within the ground-truth mask. SR is sensitive to vessel breakage and missing branches, while SP penalizes spurious or disconnected predictions. Formally, they are expressed as:

$$\text{SR} = \frac{|S_L \cap V_P|}{|S_L|}, \quad \text{SP} = \frac{|S_P \cap V_L|}{|S_P|} \quad (17)$$

To assess statistical significance, we perform paired t-tests between ACE-ProtoNet and compared methods. A p-value < 0.05 is considered statistically significant. Higher values of DSC, SE, clDice, SR, and SP, and lower ASSD values, indicate better segmentation performance.

4.3. Implementation details

All experiments were conducted on an Ubuntu 18.04 server equipped with NVIDIA RTX 3090 GPUs, using the PyTorch deep learning framework. We adopted a strict five-fold cross-validation protocol to ensure the robustness and reliability of the evaluation. In each fold, one subset was used for validation while the remaining four subsets were used for training. To guarantee strictly fair comparisons, all baseline methods were re-implemented and trained from scratch under identical data splits and preprocessing configurations. The final performance was reported as the average quantitative segmentation result across the five folds.

For preprocessing, we performed intensity normalization to enhance contrast and suppress irrelevant information. Specifically, voxel intensities were truncated to a dataset-specific range: [-250, 450] HU for CTA119, ASOCA, and ImageCAS; [-60, 140] HU for PVS203; [-160, 160] HU for MIDAS; and [-1000, 400] HU for ATM22, and then linearly mapped to the range [0, 255].

During training, we randomly cropped 3D sub-volumes of size $128 \times 160 \times 160$ as model input and employed on-the-fly data augmentation, including random rotations, flipping, and Gaussian blurring. For network initialization, the VFM encoder was initialized with pretrained SAM-Med3D weights⁵, with the final two ViT blocks unfrozen for

fine-tuning, while the CNN encoder and all other new layers used He normal initialization. The model was trained for up to 800 epochs using the Adam optimizer with an initial learning rate of 1×10^{-3} . A poly learning rate policy progressively decayed the learning rate until convergence (or until the rate dropped below 1×10^{-8}), and a batch size of 2 was used. To capture the intra-class variance of vascular structures while maintaining a compact and stable embedding space, the number of prototypes per class was set to $K = 3$ for all experiments, unless otherwise specified.

At inference time, we applied a sliding-window strategy with the same patch size and a 50% overlap in all spatial dimensions to process full-resolution volumes. The probability outputs from overlapping sub-volumes were aggregated, and a final binarized segmentation map was obtained using a threshold of 0.5. To reduce false positives, a post-processing step based on connected-component analysis was applied to the segmentation outputs. For coronary artery segmentation, the two largest components were retained to account for its two principal anatomical branches, whereas for portal venous system and cerebrovascular segmentation, only the largest connected component was preserved.

5. Results and analysis

5.1. Comparisons with the state-of-the-art methods

We compared our method against eleven state-of-the-art approaches spanning diverse architectural paradigms. Specifically, the comparison includes:

- **CNN-based methods:** Four foundational architectures that excel at modeling local image features through convolutional operations: U-Net3D (Çiçek et al., 2016), CPFNet (Feng et al., 2020), S²CA-Net (Zhou et al., 2024a), and MSDUNet (Li et al., 2025).
- **Transformer- and VFM-based models:** Four methods that leverage global self-attention for long-range dependency modeling, including MedT (Valanarasu et al., 2021), UNETR (Hatamizadeh et al., 2022), TransUNet (Chen et al., 2024b), and nnFormer (Zhou et al., 2023), as well as the representative VFM-hybrid pipeline, Ensembled-SAMs (Chen et al., 2024a).
- **Vessel-specific methods:** Three networks explicitly designed for tubular vascular structures: CS²Net (Mou et al., 2021), 3D-FFR-UNET (Song et al., 2022), and VSNet (Xu et al., 2025).

To ensure a fair comparison, we applied identical data preprocessing to all experiments and retrained all competing models using the same learning strategy to obtain their best performance. Quantitative results are reported on three benchmark datasets: CTA119, ASOCA, and ImageCAS.

5.1.1. Quantitative results

Quantitative results on the three benchmark datasets are reported in Table 2 and illustrated in the bar charts of Fig. 5. The visualizations provide an intuitive comparison, showing that ACE-ProtoNet consistently achieves superior performance and attains the best results across all six evaluation metrics on each dataset.

On the in-house CTA119 dataset, ACE-ProtoNet achieves a DSC of 87.34%, a clear improvement of 2.76% over the best-performing CNN-based model (MSDUNet at 84.58%) and 2.24% over the strongest vessel-specific method (VSNet at 85.10%). It also yields the lowest ASSD of 0.71 mm and the highest clDice of 92.79%, highlighting superior boundary accuracy and topological preservation. In terms of skeleton connectivity, ACE-ProtoNet attains an SR of 91.91% and an SP of 93.69%, significantly outperforming the second-best VSNet (SR: 90.71%, SP: 92.61%) and indicating reduced vessel fragmentation.

On the public ASOCA dataset, ACE-ProtoNet maintains strong performance with a DSC of 90.18% and an ASSD of 0.65 mm, representing

⁵ <https://github.com/uni-medical/SAM-Med3D>

⁶ https://huggingface.co/blueyo0/SAM-Med3D/blob/main/sam_med3d_turbo.pth

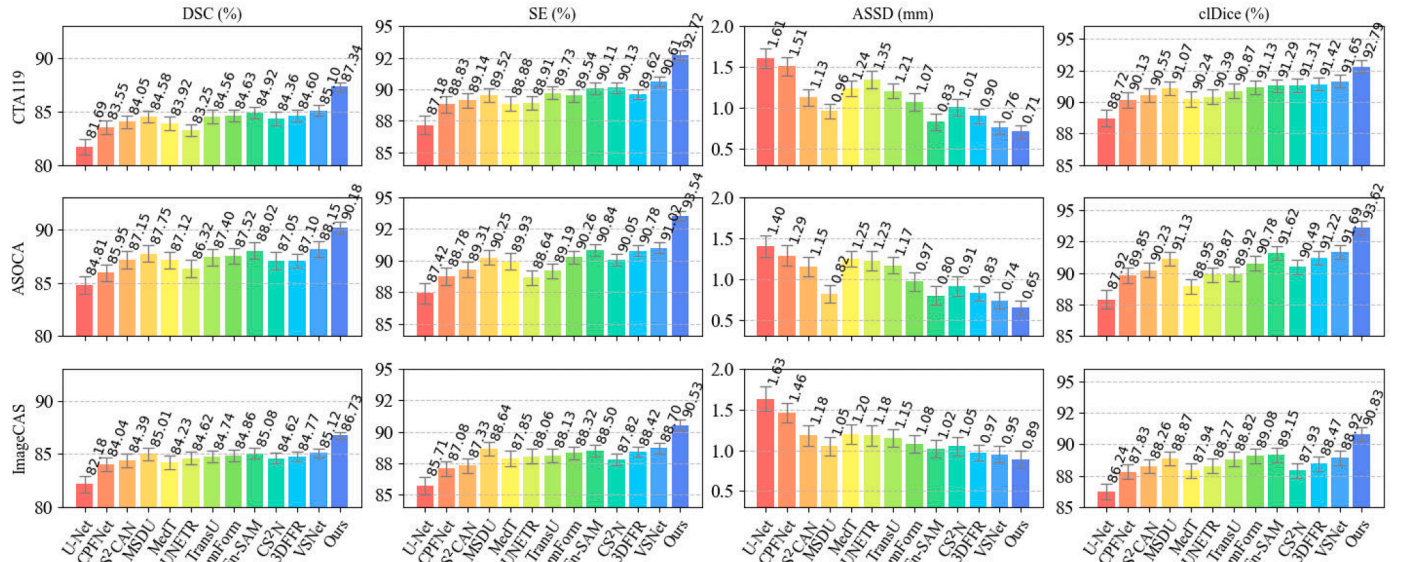


Fig. 5. Overall performance comparison of different methods. Quantitative results are reported on the CTA119, ASOCA, and ImageCAS datasets in terms of DSC (%), SE (%), ASSD (mm), and cDice (%). Higher DSC, SE, and cDice values and lower ASSD values indicate better segmentation performance. Method names are abbreviated in the figure for brevity.

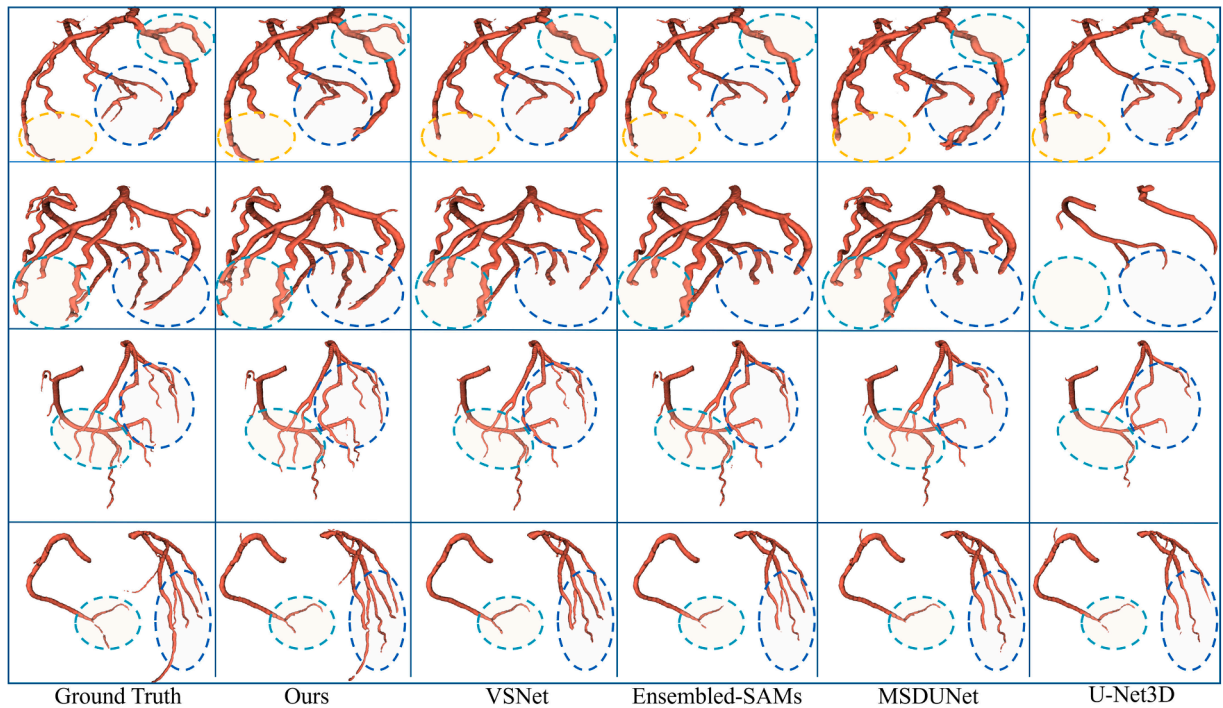


Fig. 6. Qualitative comparison of 3D coronary artery segmentation results on four representative cases. The regions highlighted by dashed circles indicate areas where competing methods exhibit significant under-segmentation or over-segmentation, leading to suboptimal segmentation results.

a 2.03% DSC gain over VSNet (88.15%). It also achieves the highest cDice (93.62%), confirming consistent generalization across both normal and diseased vessels. This advantage extends to skeleton metrics, where our method achieves an SR of 92.84% and an SP of 94.41%, ensuring the generation of clean, noise-free vascular skeletons even in complex anatomies.

For the large-scale ImageCAS dataset, which features high inter-patient variability, ACE-ProtoNet again leads with a DSC of 86.73% and a cDice of 90.83%, surpassing VSNet (85.12%, 88.92%) by a notable margin. Its ASSD of 0.89 mm, the lowest among all competitors, demonstrates robustness in challenging clinical scenarios. It also achieves the

highest SR (90.04%) and SP (91.63%), further validating its ability to maintain topological completeness.

Overall, ACE-ProtoNet consistently leads across all datasets and evaluation metrics. All improvements were found to be statistically significant ($p < .01$, paired t -test), confirming the reliability of the observed gains.

5.1.2. Qualitative results

The qualitative results in Fig. 6 and Fig. 7 further support the quantitative findings. The 3D renderings (Fig. 6) provide a holistic view of vascular topology and serve as the primary evidence of segmentation

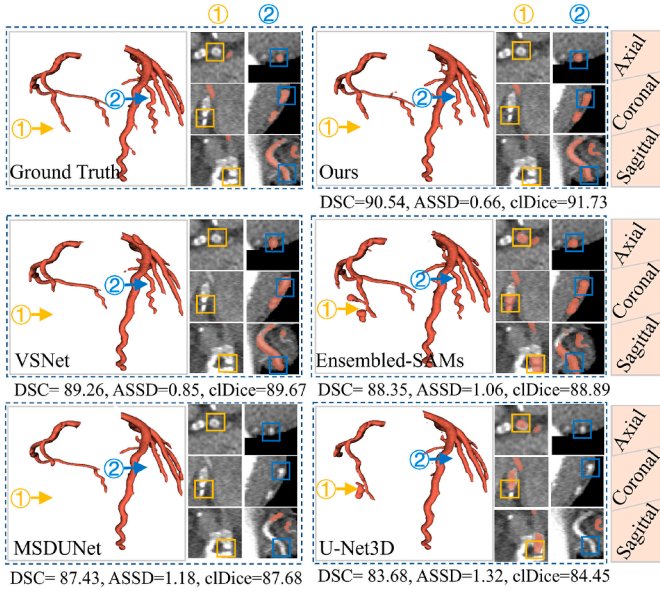


Fig. 7. Detailed 2D and 3D qualitative comparison on a challenging case, with the case-specific DSC score reported below each panel. Each subfigure displays the final 3D rendering alongside corresponding multi-planar (Axial, Coronal, Sagittal) 2D views, where segmentation results are overlaid on the original CT images. Two challenging regions of interest are highlighted: a thin distal vessel (arrow ①, yellow box) and a complex bifurcation (arrow ②, blue box).

Table 3

Comparison of computational complexity and efficiency on the CTA119 dataset. Parameters (M) denote the number of model parameters, FLOPs (G) represent the computational cost, memory (MB) indicates the memory consumption during execution, train-T (s/epoch) refers to the training time, and inf-T (s/volume) denotes the inference time. All metrics were evaluated with a $128 \times 160 \times 160$ input on an RTX 3090 GPU.

Methods	Params	FLOPs	Memory	Train-T	Inf-T
U-Net3D	16.32	425.2	4520	45	3.5
MSDUNet	35.84	650.4	6150	78	6.2
nnFormer	149.34	888.5	14,850	215	35.4
Ensembled-SAMs	258.41	2,145.6	21,500	450	85.6
CS ² Net	32.05	546.1	5800	85	5.8
VSNet	28.52	485.3	5200	65	4.5
Ours	55.43	682.4	10,240	158	22.4

quality, while the 2D slices (Fig. 7) validate performance in anatomically challenging regions.

In Fig. 6, foundational architectures such as U-Net3D exhibit frequent topological discontinuities, leading to missed coronary artery segments (dashed circles). More advanced CNN-based models (e.g., MSDUNet) and vessel-specific methods (e.g., VSNet) better capture the main trunks but still under-segment thin distal branches and produce incomplete bifurcations. Transformer-based and VFM-driven approaches (e.g., Ensembled-SAMs) improve global contextual reasoning, yet tend to over-smooth small vessels, resulting in the loss of fine-grained detail. These issues are further evidenced in Fig. 7, where the yellow (①) and blue (②) boxes highlight distal branch and bifurcation errors.

By contrast, ACE-ProtoNet delivers continuous and anatomically coherent 3D reconstructions (Fig. 6), avoiding the fragmentation and detail loss seen in prior methods. The 2D slices (Fig. 7) further corroborate its ability to accurately delineate thin vessels and complex bifurcations. Together, these results underscore ACE-ProtoNet's ability to jointly preserve global topology and local detail, establishing a new state-of-the-art for coronary artery segmentation.

In summary, ACE-ProtoNet consistently surpasses existing methods across diverse datasets and evaluation metrics. Its improvements are not only statistically significant but also clinically relevant, particularly for reliable lumen quantification, stenosis detection, and surgical planning. These findings validate the effectiveness of the proposed architecture in tackling two persistent challenges in coronary segmentation: (1) effectively integrating heterogeneous features without redundancy, and (2) delineating ambiguous or low-contrast vessels under uncertainty. This dual strength establishes ACE-ProtoNet as a robust and clinically viable solution for coronary artery segmentation.

5.1.3. Computational complexity analysis

To further evaluate the computational efficiency and deployment feasibility of the proposed method, we conducted a systematic complexity analysis under the identical experimental settings described in Section 4.3. All measurements were performed on a single NVIDIA RTX 3090 GPU. The evaluation metrics include the number of parameters (Params), floating-point operations (FLOPs), GPU memory consumption during training, average training time per epoch, and average inference time per full-resolution CT volume.

As summarized in Table 3, ACE-ProtoNet exhibits a moderate increase in parameter size and computational cost compared with CNN-based architectures. This additional overhead primarily arises from the incorporation of the pretrained VFM encoder and the uncertainty-aware prototype learning module. To control computational cost, we fine-tuned only the final two ViT blocks of the 12-layer SAM-Med3D encoder while keeping the remaining layers frozen, which effectively limits the number of trainable parameters and stabilizes memory usage during backpropagation. Compared with more computationally intensive Transformer-based models and ensemble architectures (e.g., Ensembled-SAMs), the proposed method maintains a relatively lower memory footprint while preserving efficient inference performance. The average inference time is approximately 22.4 seconds per full-resolution CT volume, demonstrating practical feasibility for routine computer-aided clinical analysis workflows.

5.2. Ablation studies

To validate the effectiveness of the proposed method, we conducted ablation experiments focusing on: (1) the contribution of each key component, (2) the impact of loss parameter configurations, and (3) the sensitivity to the number of prototypes.

5.2.1. Effectiveness of key components

To systematically validate the contributions of individual components, we conducted a comprehensive ablation study on the CTA119, ASOCA, and ImageCAS datasets. Our baseline model (Model1) is a standard CNN encoder-decoder, similar to the one adopted in Dong et al. (2024). Starting from this baseline, we progressively integrated our proposed modules to evaluate their distinct and synergistic impacts. Improvements were assessed using both quantitative metrics (Table 4) and qualitative visualizations (Fig. 8).

Benefit of the dual-encoder Replacing the single encoder with a dual-encoder backbone incorporating feature summation (Model2) leads to consistent performance gains. Specifically, the DSC improves from 82.89% to 84.64% on CTA119 and from 85.24% to 86.43% on ASOCA, accompanied by ASSD reductions from 1.42 mm to 1.17 mm and from 1.39 mm to 1.13 mm, respectively. These results demonstrate the benefit of integrating the VFM's global structural representations with the CNN's fine-grained local features. However, the diffuse activation map in Fig. 8(c) suggests that a more advanced fusion mechanism is desirable.

Impact of the ACE-Gate Substituting simple feature summation with the proposed ACE-Gate (Model3) delivers another significant performance leap. The DSC increases to 86.28% on CTA119 and 88.31% on

Table 4

Step-by-step ablation study on the CTA119, ASOCA, and ImageCAS datasets. The baseline is a pure CNN encoder-decoder. Subsequent variants progressively add the key components of our proposed ACE-ProtoNet. Best results are highlighted in **bold**.

Net	Base-line	VFM	UPL-Head	CTA119	ASOCA					ImageCAS				
					DSC(%)↑	SE(%)↑	ASSD(mm)↓	clDice(%)↑	DSC(%)↑	SE(%)↑	ASSD(mm)↓	clDice(%)↑	DSC(%)↑	SE(%)↑
Model1	✓	-	-	-	82.89±0.54	89.00±0.38	1.42±0.11	89.90±0.68	85.24±0.60	89.38±0.45	1.39±0.10	89.80±0.70	82.56±0.35	88.50±0.44
Model2	✓	✓	-	-	84.64±0.49	90.13±0.35	1.17±0.10	91.02±0.62	86.43±0.59	91.12±0.43	1.13±0.08	91.15±0.66	84.38±0.36	89.37±0.38
Model3	✓	-	-	✓	86.28±0.47	91.36±0.37	0.91±0.09	92.30±0.59	88.31±0.59	92.09±0.43	0.86±0.10	92.40±0.63	85.15±0.34	90.06±0.45
Model4	✓	-	✓	-	85.12±0.48	90.07±0.36	1.14±0.11	91.20±0.60	86.70±0.38	91.00±0.44	1.08±0.10	91.21±0.68	84.62±0.35	89.30±0.36
Ours	✓	-	✓	✓	87.34±0.43	92.72±0.36	0.71±0.08	92.79±0.51	90.18±0.57	93.54±0.42	0.65±0.09	93.62±0.53	86.73±0.34	90.53±0.41

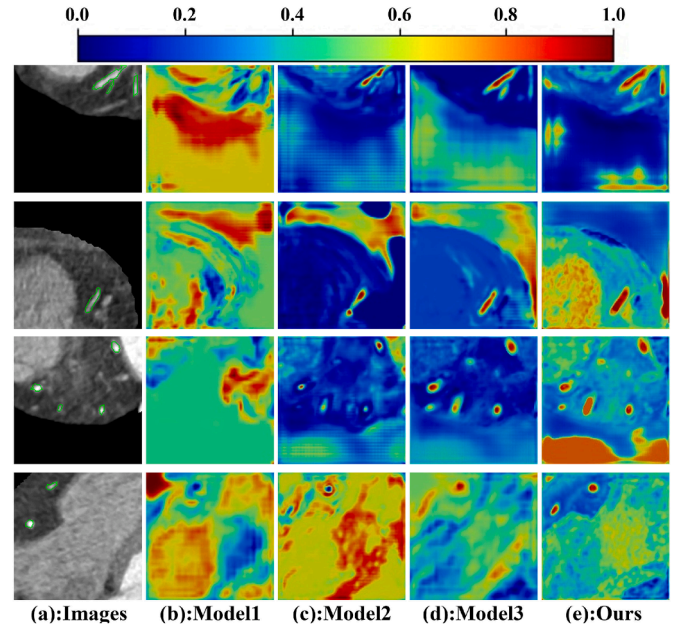


Fig. 8. Visual ablation study of model components via feature activation maps. Columns display: (a) The original image with ground-truth (green); (b) baseline (Model1); (c) Model1 + VFM (using Sum fusion, Model2); (d) Model1 + VFM + ACE-Gate (Model3); and (e) the full model with UPL-Head (Ours). Heatmaps from the encoder's final layer indicate model attention (warmer colors are higher). The progressive sharpening of the activation maps from (b) to (e) visually confirms the efficacy of each added component and their synergy in the final model.

Table 5

Ablation study of the voxel-prototype contrastive loss weight β on the CTA119 dataset. Best results are highlighted in **bold**.

β	DSC (%)	SE (%)	ASSD (mm)	clDice (%)
0.0	86.81 ± 0.46	91.75 ± 0.38	0.97 ± 0.10	92.07 ± 0.57
0.1	87.06 ± 0.44	92.18 ± 0.39	0.86 ± 0.09	92.46 ± 0.54
0.2	87.34 ± 0.43	92.72 ± 0.36	0.71 ± 0.08	92.79 ± 0.51
0.3	86.88 ± 0.44	91.94 ± 0.39	0.94 ± 0.09	92.33 ± 0.53
0.4	86.65 ± 0.43	91.33 ± 0.36	1.12 ± 0.10	91.88 ± 0.54
0.5	86.12 ± 0.45	90.76 ± 0.37	1.23 ± 0.11	91.15 ± 0.55

ASOCA, while the ASSD decreases to 0.91 mm and 0.86 mm, respectively. As depicted in Fig. 8(d), the activation maps become sharper and more localized, confirming that the ACE-Gate effectively reconciles heterogeneous representations through its uncertainty-aware adaptive fusion.

Contribution of the UPL-Head Serving as the key refinement module, the UPL-Head consistently enhances segmentation quality across all datasets. When incorporated into the baseline (Model4), it independently enhances the DSC across all datasets (e.g., from 82.89% to 85.12% on CTA119). Its synergistic effect is most evident in the full model (Ours), where performance peaks with DSCs of 87.34% (CTA119), 90.18% (ASOCA), and 86.73% (ImageCAS). As illustrated in Fig. 8(e), the uncertainty-aware prototype learning yields the most anatomically faithful segmentations, capturing fine-grained vascular details with high precision.

In conclusion, the complete ACE-ProtoNet, integrating all proposed components, achieves the best overall performance across all benchmarks, with top DSC scores of 87.34%, 90.18%, and 86.73% on CTA119, ASOCA, and ImageCAS, respectively. The consistent stepwise improvements in both quantitative metrics and qualitative visualizations demonstrate that each component contributes uniquely yet synergistically, forming a robust and anatomically faithful framework for coronary artery segmentation.

Table 6

Ablation study on the number of prototypes K on the CTA119 dataset. The best results are highlighted in **bold**.

K	DSC (%) $\uparrow$	SE (%) $\uparrow$	ASSD (mm) $\downarrow$	clDice (%) $\uparrow$
1	86.15 $\pm$ 0.52	91.20 $\pm$ 0.45	0.98 $\pm$ 0.12	91.45 $\pm$ 0.65
2	86.92 $\pm$ 0.48	92.15 $\pm$ 0.40	0.82 $\pm$ 0.10	92.10 $\pm$ 0.58
3	87.34 $\pm$ 0.43	92.72 $\pm$ 0.36	0.71 $\pm$ 0.08	92.79 $\pm$ 0.51
5	87.09 $\pm$ 0.45	92.48 $\pm$ 0.38	0.76 $\pm$ 0.09	92.55 $\pm$ 0.55
7	86.88 $\pm$ 0.50	92.05 $\pm$ 0.42	0.85 $\pm$ 0.11	92.25 $\pm$ 0.60

5.2.2. Effect of the voxel-prototype contrastive loss

We assessed the influence of the voxel-prototype contrastive loss ($\mathcal{L}_{ppc}$) on model performance by varying its weight $\beta \in \{0.0, 0.1, \dots, 0.5\}$, with results detailed in Table 5. Without this loss ($\beta = 0.0$), the model relies solely on voxel-level supervision and yields the lowest DSC of 86.81%, indicating that the absence of embedding regularization limits feature discriminability. Performance improves as β increases, reaching its optimum at $\beta = 0.2$ with a DSC of 87.34% and an ASSD of 0.71 mm. This moderate weight strikes the best trade-off, enhancing intra-class compactness and inter-class separability without compromising voxel-level precision. Further increases in β (≥ 0.3) lead to a slight performance decline, suggesting that excessive weighting on $\mathcal{L}_{ppc}$ can overshadow fine-grained supervision and result in overly compact feature clusters. Therefore, we establish $\beta = 0.2$ as the optimal setting for achieving a highly discriminative feature space and superior segmentation accuracy.

5.2.3. Impact of the number of prototypes (K)

To investigate the impact of the number of prototypes per class, we evaluated the proposed framework on the CTA119 dataset with $K \in \{1, 2, 3, 5, 7\}$. A single prototype ($K = 1$) may be insufficient to capture the rich intra-class variability of coronary artery features, which arise from variations in vessel diameter, curvature, and branching patterns. Introducing a small number of prototypes allows the model to better represent these diverse feature patterns while maintaining a compact and stable embedding space.

As shown in Table 6, segmentation performance consistently improves as K increases from 1 to 3. The proposed framework achieves the best overall performance at $K = 3$, obtaining a DSC of 87.34% and an ASSD of 0.71 mm. Further increasing K to 5 or 7 leads to a slight performance degradation. Overall, the relatively small performance variation across different values of K indicates that the proposed uncertainty-aware prototype learning mechanism is robust to this hyperparameter.

Notably, the performance degradation observed at $K \geq 5$ suggests that the representational capacity of the model saturates at relatively small prototype numbers. A key limitation relates to data efficiency. When the total number of training voxels is fixed, the effective sample support per prototype scales approximately as N/K , meaning that excessively large K values lead to fewer samples per prototype and less stable estimation.

In addition, coronary artery structures exhibit relatively low intrinsic anatomical variability compared with large-scale natural image datasets. Their primary intra-class variations can be effectively captured using a compact prototype basis. Increasing K beyond this intrinsic variability results in unnecessary partitioning of the embedding space, without further improving representation quality. Therefore, a compact configuration (e.g., $K = 3$) provides a favorable balance between representational expressiveness and statistical stability. This interpretation is consistent with the empirical saturation observed in Table 6.

6. Discussion

In this section, we carry out more experiments to demonstrate and discuss the superiority of our design.

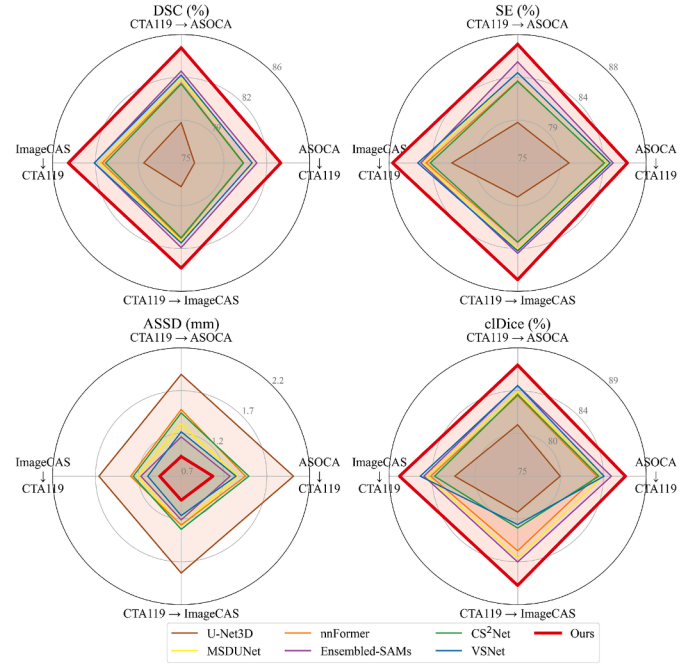


Fig. 9. Radar charts comparing the cross-domain generalization performance of our proposed method against six state-of-the-art methods in scenarios between the CCTA119 dataset and the ASOCA and ImageCAS datasets. The charts show results for DSC, SE, ASSD, and clDice across the four resulting training-testing scenarios. Our method (ours, bold red line) achieves the best performance, indicated by the largest area for the first three metrics and the smallest for ASSD.

6.1. Generalization to unseen domains

A persistent challenge for deep learning models in medical imaging is their vulnerability to domain shifts, which arise from variations in scanner protocols, patient populations, and acquisition parameters across institutions. To evaluate the robustness of ACE-ProtoNet against such shifts, we conducted a comprehensive cross-domain generalization study, training the model on one dataset and directly testing it on an entirely unseen dataset without any fine-tuning.

The quantitative outcomes, presented in Table 7 and summarized in the radar plots of Fig. 9, confirm ACE-ProtoNet's state-of-the-art generalization capability. Across all four domain-shift scenarios (CTA119 $\leftrightarrow$ ASOCA and CTA119 $\leftrightarrow$ ImageCAS), our model consistently outperforms the six competing methods. For example, in the CTA119 $\rightarrow$ ASOCA setting, ACE-ProtoNet achieves a DSC score of 84.86%, surpassing the next-best method (Ensembled-SAMs) by +1.99%. This advantage extends to other metrics, with a +1.77% gain in sensitivity and a $\sim 20\%$ relative reduction in surface distance error (ASSD: 0.93 mm vs. 1.16 mm). Similar superiority is observed in the ImageCAS $\rightarrow$ CTA119 scenario, where our model again leads in DSC (84.65%) and achieves the lowest ASSD (0.95 mm). These consistent gains underscore the ability of ACE-ProtoNet to preserve vascular topology and capture fine-grained details under significant distribution shifts.

Qualitative results further illustrate the model's behavior under domain shifts. As shown in Fig. 10, cross-domain predictions are contrasted with in-domain counterparts, which serve as a practical reference. In the challenging ASOCA $\rightarrow$ CTA119 (Fig. 10C) and ImageCAS $\rightarrow$ CTA119 (Fig. 10G) scenarios, the domain gap leads to some under-segmentation of distal branches (blue). Nevertheless, ACE-ProtoNet effectively preserves the connectivity of major arteries and reconstructs anatomically faithful vessel structures. Moreover, models trained on CTA119 tend to generate richer and more complete vessel trees, including vessels visible in CCTA but not annotated in the reference standard (Fig. 10A). This reflects differences in annotation granularity: CTA119 contains 119

Table 7

Quantitative results for the cross-domain generalization experiments between the CCTA119 dataset and the ASOCA and ImageCAS datasets. Best results are highlighted in **bold**.

Method	CTA119 → ASOCA				ASOCA → CTA119			
	DSC (%)	SE (%)	ASSD (mm)	clDice (%)	DSC (%)	SE (%)	ASSD (mm)	clDice (%)
U-Net3D(Çiçek et al., 2016)	78.45 ± 0.47	79.10 ± 0.50	1.89 ± 0.10	80.62 ± 0.56	76.14 ± 0.48	80.20 ± 0.51	2.01 ± 0.08	79.64 ± 0.59
MSDUNet(Li et al., 2025)	82.24 ± 0.46	84.16 ± 0.46	1.31 ± 0.07	84.23 ± 0.53	81.13 ± 0.45	84.16 ± 0.48	1.36 ± 0.06	84.31 ± 0.56
nnFormer(Zhou et al., 2023)	81.85 ± 0.42	83.31 ± 0.44	1.48 ± 0.06	83.79 ± 0.50	80.35 ± 0.44	83.72 ± 0.46	1.45 ± 0.05	83.61 ± 0.52
Ensembled-SAMs(Chen et al., 2024a)	82.87 ± 0.38	85.24 ± 0.41	1.16 ± 0.05	84.90 ± 0.48	81.48 ± 0.41	84.66 ± 0.43	1.27 ± 0.04	85.21 ± 0.44
CS ² Net(Mou et al., 2021)	81.76 ± 0.43	83.25 ± 0.45	1.44 ± 0.07	83.93 ± 0.52	80.31 ± 0.43	83.78 ± 0.46	1.49 ± 0.05	83.84 ± 0.51
VSNet(Xu et al., 2025)	82.53 ± 0.44	84.14 ± 0.45	1.22 ± 0.06	84.88 ± 0.51	81.07 ± 0.45	84.35 ± 0.47	1.34 ± 0.05	84.41 ± 0.53
Ours	84.86 ± 0.37	87.01 ± 0.40	0.93 ± 0.03	87.12 ± 0.42	83.52 ± 0.41	86.10 ± 0.44	1.07 ± 0.06	86.73 ± 0.47

Method	CTA119 → ImageCAS				ImageCAS → CTA119			
	DSC (%)	SE (%)	ASSD (mm)	clDice (%)	DSC (%)	SE (%)	ASSD (mm)	clDice (%)
U-Net3D(Çiçek et al., 2016)	77.03 ± 0.52	78.44 ± 0.49	1.83 ± 0.08	78.91 ± 0.60	78.20 ± 0.48	81.66 ± 0.47	1.66 ± 0.07	81.85 ± 0.57
MSDUNet(Li et al., 2025)	81.71 ± 0.47	83.67 ± 0.45	1.28 ± 0.06	83.92 ± 0.57	82.03 ± 0.45	84.55 ± 0.46	1.21 ± 0.05	84.86 ± 0.54
nnFormer(Zhou et al., 2023)	81.47 ± 0.43	83.05 ± 0.44	1.27 ± 0.05	83.13 ± 0.54	81.76 ± 0.42	84.25 ± 0.43	1.29 ± 0.05	84.47 ± 0.52
Ensembled-SAMs(Chen et al., 2024a)	82.25 ± 0.39	84.15 ± 0.41	1.21 ± 0.04	84.35 ± 0.45	82.43 ± 0.37	84.79 ± 0.39	1.17 ± 0.03	85.21 ± 0.43
CS ² Net(Mou et al., 2021)	81.40 ± 0.44	83.01 ± 0.45	1.32 ± 0.06	80.65 ± 0.51	81.47 ± 0.42	83.83 ± 0.44	1.26 ± 0.05	84.05 ± 0.50
VSNet(Xu et al., 2025)	81.85 ± 0.46	83.90 ± 0.45	1.16 ± 0.05	80.27 ± 0.55	82.44 ± 0.43	85.10 ± 0.44	1.09 ± 0.05	85.61 ± 0.52
Ours	84.01 ± 0.38	86.82 ± 0.41	0.98 ± 0.03	86.90 ± 0.43	84.65 ± 0.35	87.64 ± 0.39	0.95 ± 0.03	87.86 ± 0.40

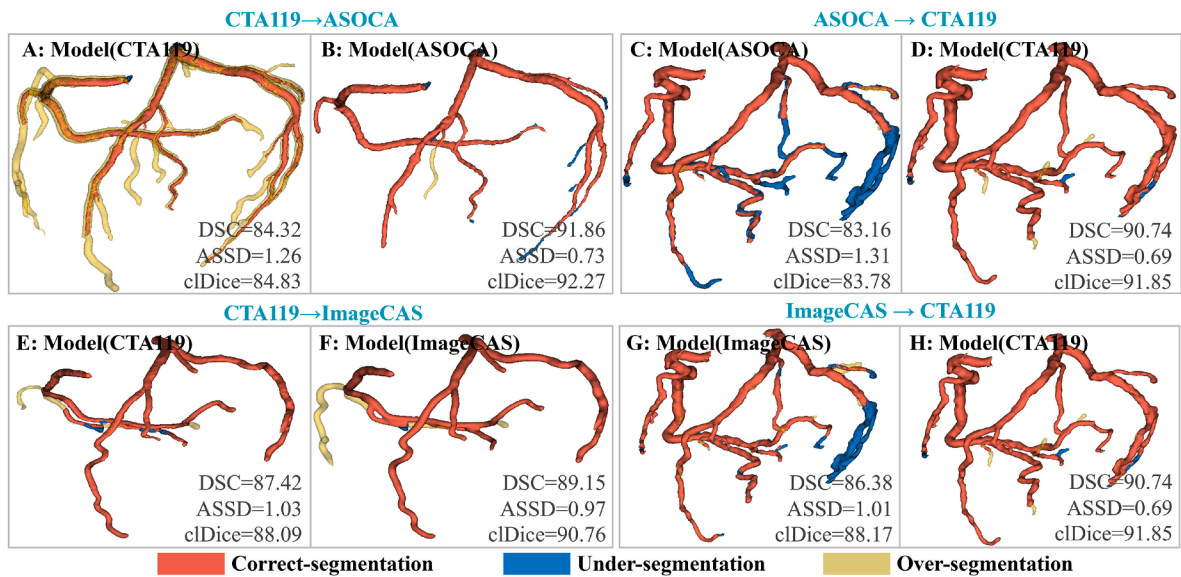


Fig. 10. Cross-domain validation results between the CCTA119 dataset and the ASOCA and ImageCAS datasets across four training-testing configurations. In the CTA119→ASOCA setting, (A) denotes training on CTA119 and testing on ASOCA, while (B) denotes in-domain training and testing on ASOCA. Similar configurations apply to ASOCA→CTA119 (C-D), CTA119→ImageCAS (E-F), and ImageCAS→CTA119 (G-H). DSC, ASSD, and clDice are reported for each case (higher DSC/clDice and lower ASSD indicate better performance). Correctly segmented regions are shown in red, under-segmentation in blue, and over-segmentation in yellow.

cases with detailed SCCT coronary artery segmentation criteria, whereas ASOCA provides only 40 cases with simpler annotations focused mainly on primary branches.

Overall, ACE-ProtoNet demonstrates robust generalization to domain shifts and consistent performance across datasets with varying annotation complexities, underscoring its suitability for real-world CAD applications.

6.2. Transferability across anatomy and modality

To further assess the transferability and robustness of the proposed architecture beyond coronary artery segmentation, we evaluated its performance on three representative segmentation tasks: portal venous system segmentation from abdominal CT (PVS203), cerebrovascular segmentation from MRA (MIDAS), and pulmonary airway segmentation from thoracic CT (ATM22). For fair comparison, all competing methods

were retrained from scratch on each dataset under identical experimental settings.

6.2.1. Cross-anatomy transferability: PVS203 (abdominal portal veins)

The PVS203 dataset represents a markedly different anatomical domain, characterized by distinct vascular morphologies and branching patterns compared to coronary arteries. As reported in Table 8, ACE-ProtoNet exhibits strong cross-anatomy transferability. It achieves a DSC of 92.55%, outperforming the next-best method (VSNet) by 2.39%, and consistently surpasses other methods across additional quantitative metrics. All competing methods were retrained under identical settings to ensure a fair comparison.

Qualitative visualizations in Fig. 11 further support these findings. While several competing methods produce fragmented or discontinuous segmentations, ACE-ProtoNet reconstructs an anatomically coherent and topologically complete portal venous tree that closely aligns with

Table 8

ACE-ProtoNet performance on cross-anatomy and cross-modality segmentation tasks compared to state-of-the-art methods. Best results are in **bold**.

Dataset	Method	DSC (%) $\uparrow$	SE (%) $\uparrow$	ASSD (mm) $\downarrow$	clDice (%) $\uparrow$
PVS203	MSDUNet(Li et al., 2025)	89.74 $\pm$ 0.54	90.39 $\pm$ 0.57	1.36 $\pm$ 0.12	90.66 $\pm$ 0.62
	nnFormer(Zhou et al., 2023)	88.77 $\pm$ 0.50	89.59 $\pm$ 0.55	1.48 $\pm$ 0.11	90.06 $\pm$ 0.59
	Ensembled-SAMs(Chen et al., 2024a)	90.11 $\pm$ 0.47	90.78 $\pm$ 0.51	1.38 $\pm$ 0.13	91.25 $\pm$ 0.59
	CS ² Net(Mou et al., 2021)	88.97 $\pm$ 0.51	89.13 $\pm$ 0.57	1.42 $\pm$ 0.12	90.15 $\pm$ 0.61
	VSNet(Xu et al., 2025)	90.16 $\pm$ 0.44	90.80 $\pm$ 0.53	1.33 $\pm$ 0.10	91.35 $\pm$ 0.56
	Ours	92.55 $\pm$ 0.40	92.80 $\pm$ 0.46	1.15 $\pm$ 0.08	93.09 $\pm$ 0.50
MIDAS	MSDUNet(Li et al., 2025)	69.27 $\pm$ 0.70	75.02 $\pm$ 0.78	1.44 $\pm$ 0.17	79.92 $\pm$ 0.73
	nnFormer(Zhou et al., 2023)	68.05 $\pm$ 0.66	74.88 $\pm$ 0.72	1.56 $\pm$ 0.15	78.92 $\pm$ 0.69
	Ensembled-SAMs(Chen et al., 2024a)	69.82 $\pm$ 0.63	76.23 $\pm$ 0.66	1.40 $\pm$ 0.14	80.27 $\pm$ 0.70
	CS ² Net(Mou et al., 2021)	68.19 $\pm$ 0.69	74.77 $\pm$ 0.73	1.54 $\pm$ 0.18	78.82 $\pm$ 0.75
	VSNet(Xu et al., 2025)	69.88 $\pm$ 0.64	76.48 $\pm$ 0.69	1.42 $\pm$ 0.13	80.46 $\pm$ 0.66
	Ours	72.50 $\pm$ 0.58	78.21 $\pm$ 0.63	1.19 $\pm$ 0.09	82.79 $\pm$ 0.61
ATM22	MSDUNet(Li et al., 2025)	92.12 $\pm$ 0.85	92.51 $\pm$ 0.91	1.45 $\pm$ 0.09	93.02 $\pm$ 0.92
	nnFormer(Zhou et al., 2023)	91.80 $\pm$ 0.88	92.32 $\pm$ 0.87	1.48 $\pm$ 0.10	92.70 $\pm$ 0.95
	Ensembled-SAMs(Chen et al., 2024a)	92.05 $\pm$ 0.80	92.75 $\pm$ 0.84	1.46 $\pm$ 0.08	92.98 $\pm$ 0.90
	CS ² Net(Mou et al., 2021)	91.50 $\pm$ 0.92	92.20 $\pm$ 0.93	1.50 $\pm$ 0.10	92.61 $\pm$ 1.05
	VSNet(Xu et al., 2025)	91.95 $\pm$ 0.82	92.65 $\pm$ 0.89	1.47 $\pm$ 0.09	92.84 $\pm$ 0.96
	Ours	93.23 $\pm$ 0.77	93.57 $\pm$ 0.82	1.31 $\pm$ 0.08	93.65 $\pm$ 0.85

P-values between Ours and each comparison method: <0.01 (DSC), <0.01 (SE), <0.01 (ASSD), <0.01 (clDice).

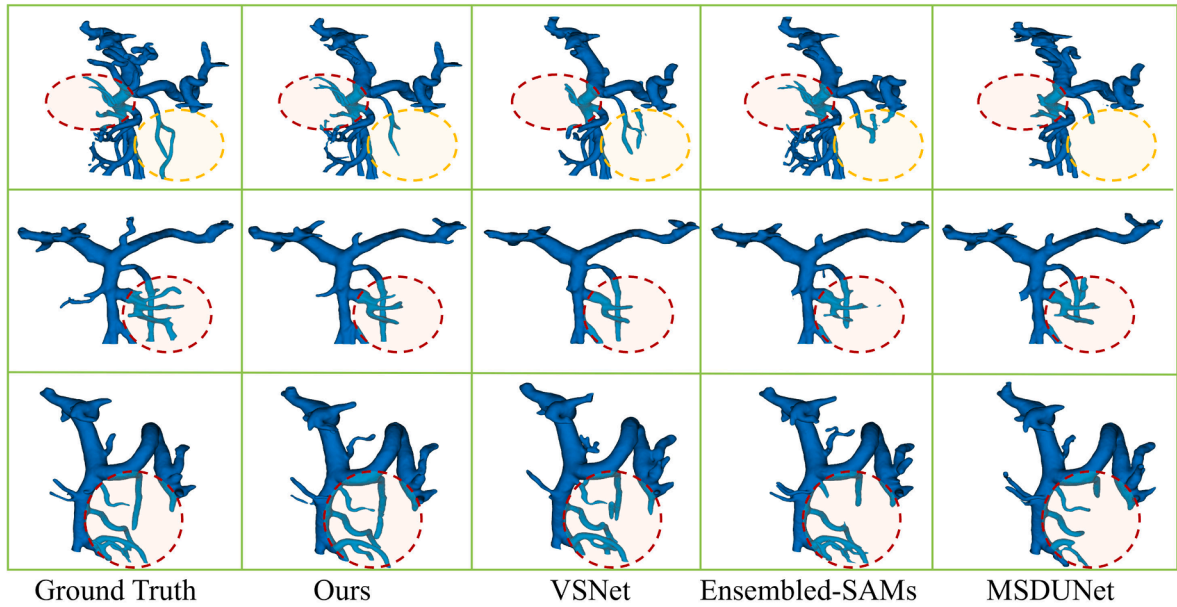


Fig. 11. 3D visualization of vessel segmentation results on the PVS203 dataset. Dashed circles denote regions of interest where competing methods may exhibit under- or over-segmentation.

the ground truth. These results highlight the model's robust performance and reliable transferability across distinct vascular anatomies.

6.2.2. Cross-modality and cross-anatomy transferability: MIDAS (cerebrovascular MRA)

To evaluate ACE-ProtoNet's robustness under simultaneous variations in anatomy and imaging modality, we benchmarked all methods on the MIDAS MRA dataset. Despite the substantial domain gap, ACE-ProtoNet again achieved the best overall performance, reaching a DSC of 72.50%. It outperforms the next-best method (VSNet) by 2.62% and consistently achieves superior results in SE, ASSD, and clDice (Table 8). All competing methods were retrained from scratch under identical settings for a fair comparison.

Fig. 12 illustrates the signed distance error maps, where green denotes accurate segmentation. Competing methods such as MSDUNet show over-segmentation (red regions), while VSNet tends to miss small

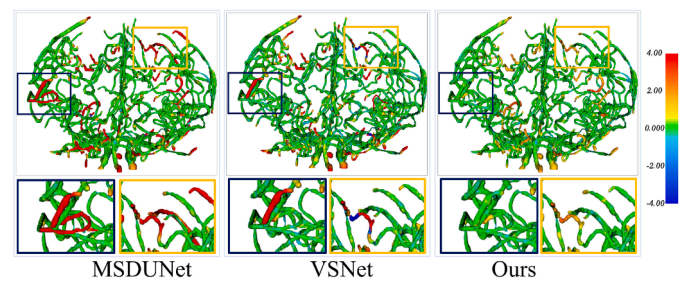


Fig. 12. 3D visualization of vessel segmentation results on the MIDAS dataset. The color bar indicates the signed distance field between the predicted surfaces and the ground-truth annotations. Warm colors (red) denote positive errors (over-segmentation) and cool colors (blue) denote negative errors (under-segmentation).

and peripheral vessels (blue regions). In contrast, ACE-ProtoNet produces a vessel tree predominantly covered in green, indicating minimal deviation from the ground truth and superior geometric fidelity.

Collectively, these results demonstrate that ACE-ProtoNet is not merely a task-specific solution, but a robust and versatile architecture capable of consistently delivering strong performance across diverse vascular anatomies and imaging modalities.

6.2.3. Broader applicability to other tubular structures

While the primary focus of this study is coronary artery segmentation, the architectural design of ACE-ProtoNet is well suited to addressing challenges that are also encountered in other tubular anatomical structures. These challenges include complex, tree-like branching topologies, pronounced scale variations between proximal trunks and distal branches, and ambiguous boundaries in low-contrast regions.

The effectiveness of this design is supported by the experimental results on multiple anatomically and modally distinct datasets. As reported in Sections 6.2.1 and 6.2.2, ACE-ProtoNet achieves competitive performance on the PVS203 dataset for portal venous system segmentation from CT and on the MIDAS dataset for cerebrovascular segmentation from MRA, both of which differ substantially from coronary arteries in terms of anatomical characteristics and imaging modalities.

Building upon these findings, we further evaluated ACE-ProtoNet on the publicly available ATM22 dataset for pulmonary airway segmentation. The pulmonary airway tree comprises the trachea and its hierarchical bronchial branches, providing a representative testbed for pulmonary tree-like structure segmentation. As summarized in Table 8, ACE-ProtoNet achieves superior performance on ATM22 in terms of DSC, ASSD, and cIDice when retrained from scratch under standard training protocols. This result is notable given that airway segmentation shares key challenges with vascular segmentation, particularly in preserving connectivity from the central trachea to fine distal branches that often exhibit weak boundaries (Tan et al., 2021).

Overall, these results indicate that ACE-ProtoNet constitutes a robust architectural framework for tubular structure segmentation, capable of accommodating diverse anatomical configurations and imaging modalities through principled feature gating and uncertainty-guided representation learning.

6.3. Limitations and future work

Despite its state-of-the-art performance and robust generalization, ACE-ProtoNet's primary limitation lies in preserving vascular topology. In some cases, the model may produce topologically incomplete segmentations, such as discontinuous vessels. The current framework also lacks a dedicated mechanism for post-hoc correction of these structural discontinuities.

Future research will focus on explicit topological rehabilitation. One promising direction, inspired by recent advances in vessel reconnection (Qiu et al., 2025), involves a multi-stage refinement strategy. This could incorporate a centerline-based stage to bridge gaps between fragmented vessels, followed by an Implicit Neural Representation (INR) module to reconstruct a continuous luminal geometry.

Additionally, we plan to integrate graph-based reasoning, as in VSR-Net (Ye et al., 2025). Disconnected vessel fragments can be modeled as graph nodes, enabling a Graph Convolutional Network (GCN) to learn their spatial and morphological relationships. Such graph-level analysis can provide high-level guidance for identifying anatomically consistent reconstructions.

By coupling these structure-aware rehabilitation techniques with our current segmentation framework, we aim to significantly improve topological completeness and move closer to clinically trustworthy coronary artery extraction.

7. Conclusion

In this paper, we presented ACE-ProtoNet, a unified framework designed to overcome the long-standing challenges of accuracy, robustness, and generalization in coronary artery segmentation. The framework integrates three synergistic innovations: (1) a parallel dual-encoder backbone that couples global structural representations from a partially frozen VFM with fine-grained local details captured by a trainable CNN; (2) the ACE-Gate, which performs statistically principled feature fusion via covariance and eigenvalue analysis; and (3) the UPL-Head, which leverages predictive uncertainty to guide prototype adaptation and enhance reliability in ambiguous vascular regions.

Extensive experiments across two in-house and three public datasets demonstrated the superior accuracy and generalization of ACE-ProtoNet, with notable advantages in delineating small, distal vessels and maintaining robustness under domain shifts. Ablation studies confirmed the effectiveness of the dual-encoder, ACE-Gate, and UPL-Head, while cross-anatomy transfer experiments further demonstrated the framework's adaptability to other vascular structures such as the portal vein and cerebrovasculature.

Future work will focus on explicit topological rehabilitation, potentially integrating graph-based reasoning to identify vessel discontinuities and multi-stage geometric refinement to ensure vascular continuity. By uniting foundation model representations with statistically principled fusion and uncertainty-guided prototype learning, ACE-ProtoNet establishes a statistically informed and clinically grounded paradigm for automated coronary artery segmentation, paving the way toward more reliable quantitative analysis and decision support in cardiovascular imaging.

CRedit authorship contribution statement

Caixia Dong: Writing – original draft, Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Funding acquisition; **Duwei Dai:** Writing – original draft, Visualization, Software, Formal analysis, Data curation, Funding acquisition; **Pengyu Ren:** Resources, Investigation, Data curation; **Guowei Dai:** Visualization, Validation, Software; **Yu Wang:** Investigation, Formal analysis, Data curation; **Linyun Zhou:** Formal analysis, Data curation; **Yang Li:** Writing – original draft, Investigation, Conceptualization, Funding acquisition; **Wei Zeng:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Abdi, H., Williams, L.J., 2010. Principal component analysis. *Wiley Interdiscip. Rev. Comput. Stat.* 2 (4), 433–459.
- Chen, F., Ge, J., Zheng, Y., Guo, K., Cao, F., Liang, J., 2024a. Ensembled-SAMs for enhanced small coronary artery segmentation in CCTA images. *IEEE J. Biomed. Health Inform.* .
- Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y., 2021. Transunet: transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*.

- Chen, J., Mei, J., Li, X., Lu, Y., Yu, Q., Wei, Q., Luo, X., Xie, Y., Adeli, E., Wang, Y., et al., 2024b. TransUNet: rethinking the u-net architecture design for medical image segmentation through the lens of transformers. *Med Image Anal.* 97, 103280.
- Çiçek, Ö., Abdulkadir, A., Lienkamp, S.S., Brox, T., Ronneberger, O., 2016. 3D u-net: learning dense volumetric segmentation from sparse annotation. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, pp. 424–432.
- Dong, C., Dai, D., Han, X., Liu, F., Yang, X., Li, Z., Xu, S., 2025. Unleashing vision foundation models for coronary artery segmentation: parallel vit-CNN encoding and variational fusion. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, pp. 647–657.
- Dong, C., Dai, D., Li, Z., Xu, S., 2024. A novel deep network with triangular-star spatial-spectral fusion encoding and entropy-aware double decoding for coronary artery segmentation. *Information Fusion* 112, 102561.
- Dong, C., Xu, S., Dai, D., Zhang, Y., Zhang, C., Li, Z., 2023. A novel multi-attention, multi-scale 3d deep network for coronary artery segmentation. *Med Image Anal.* 85, 102745.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al., 2020. An image is worth 16x16 words: transformers for image recognition at scale. In: *International Conference on Learning Representations*.
- Feng, S., Zhao, H., Shi, F., Cheng, X., Wang, M., Ma, Y., Xiang, D., Zhu, W., Chen, X., 2020. Cpfnet: context pyramid fusion network for medical image segmentation. *IEEE Trans. Med. Imaging* 39 (10), 3008–3018.
- Frangi, A.F., Niessen, W.J., Vincken, K.L., Viergever, M.A., 1998. Multiscale vessel enhancement filtering. In: *Medical Image Computing and Computer-Assisted Intervention-MICCAI'98: First International Conference Cambridge, MA, USA, October 11–13, 1998 Proceedings 1*. Springer, pp. 130–137.
- Gao, Z., Xie, J., Wang, Q., Li, P., 2019. Global second-order pooling convolutional networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3024–3033.
- Gharleghi, R., Adikari, D., Ellenberger, K., Webster, M., Ellis, C., Sowmya, A., Ooi, S., Beier, S., 2023. Annotated computed tomography coronary angiogram images and associated data of normal and diseased arteries. *Sci. Data* 10 (1), 128.
- Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H.R., Xu, D., 2022. Unetr: transformers for 3D medical image segmentation. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 574–584.
- Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 7132–7141.
- Jin, Y., Pepe, A., Li, J., Gsaxner, C., Chen, Y., Puladi, B., Zhao, F.-h., Pomykala, K., Kleesiek, J., Frangi, A.F., et al., 2025. Aortic vessel tree segmentation for cardiovascular diseases treatment: status quo. *ACM Comput. Surv.* 57 (9), 1–35.
- Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.-Y., et al., 2023. Segment anything. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 4015–4026.
- Kwon, Y., Won, J.-H., Kim, B.J., Paik, M.C., 2020. Uncertainty quantification using bayesian neural networks in classification: application to biomedical image segmentation. *Comput. Stat. Data Anal.* 142, 106816.
- Li, X., Li, L., Xing, X., Liao, H., Wang, W., Dong, Q., Qin, X., Yuan, C., 2025. Msdunet: a model based on feature multi-scale and dual-input dynamic enhancement for skin lesion segmentation. *IEEE Trans. Med. Imaging*.
- Li, Z., Xia, Q., Hu, Z., Wang, W., Xu, L., Zhang, S., 2021. A deep reinforced tree-traversal agent for coronary artery centerline extraction. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, pp. 418–428.
- Lin, T.-Y., RoyChowdhury, A., Maji, S., 2015. Bilinear CNN models for fine-grained visual recognition. In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 1449–1457.
- Luo, W., Li, Y., Urtasun, R., Zemel, R., 2016. Understanding the effective receptive field in deep convolutional neural networks. *Adv. Neural. Inf. Process. Syst.* 29.
- Manniesing, R., Niessen, W., 2004. Local speed functions in level set based vessel segmentation. In: *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2004: 7th International Conference, Saint-Malo, France, September 26–29, 2004. Proceedings, Part I 7*. Springer, pp. 475–482.
- Marano, R., Rovere, G., Savino, G., Flammia, F.C., Carafa, M. R.P., Steri, L., Merlino, B., Natale, L., 2020. Ccta in the diagnosis of coronary artery disease. *Radiol. Med.* 125, 1102–1113.
- Mehrtash, A., Wells, W.M., Tempny, C.M., Abolmaesumi, P., Kapur, T., 2020. Confidence calibration and predictive uncertainty estimation for deep medical image segmentation. *IEEE Trans. Med. Imaging* 39 (12), 3868–3878.
- Moeskops, P., Wolterink, J.M., Van Der Velden, B. H.M., Gilhuijs, K. G.A., Leiner, T., Viergever, M.A., Išgum, I., 2016. Deep learning for multi-task medical image segmentation in multiple modalities. In: *Medical Image Computing and Computer-Assisted Intervention-MICCAI 2016: 19th International Conference, Athens, Greece, October 17–21, 2016, Proceedings, Part II 19*. Springer, pp. 478–486.
- Mou, L., Zhao, Y., Fu, H., Liu, Y., Cheng, J., Zheng, Y., Su, P., Yang, J., Chen, L., Frangi, A.F., et al., 2021. Cs2-net: deep learning segmentation of curvilinear structures in medical imaging. *Med Image Anal.* 67, 101874.
- Nair, T., Precup, D., Arnold, D.L., Arbel, T., 2020. Exploring uncertainty measures in deep networks for multiple sclerosis lesion detection and segmentation. *Med Image Anal.* 59, 101557.
- Pan, C., Qi, B., Zhao, G., Liu, J., Fang, C., Zhang, D., Li, J., 2022. Deep 3d vessel segmentation based on cross transformer network. In: *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, pp. 1115–1120.
- Qiu, Y., Shan, D., Wang, Y., Dong, P., Wu, D., Yang, X., Hong, Q., Shen, D., 2025. A topology-preserving three-stage framework for fully-connected coronary artery extraction. *Med Image Anal.* 103, 103578.
- Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: convolutional networks for biomedical image segmentation. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, pp. 234–241.
- Shit, S., Paetzold, J.C., Sekuboyina, A., Ezhov, I., Unger, A., Zhylka, A., Pluim, J. P.W., Bauer, U., Menze, B.H., 2021. Cldice-a novel topology-preserving loss function for tubular structure segmentation. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 16560–16569.
- Snell, J., Swersky, K., Zemel, R., 2017. Prototypical networks for few-shot learning. *Adv. Neural. Inf. Process. Syst.* 30.
- Song, A., Xu, L., Wang, L., Wang, B., Yang, X., Xu, B., Yang, B., Greenwald, S.E., 2022. Automatic coronary artery segmentation of CCTA images with an efficient feature-fusion-and-rectification 3d-UNet. *IEEE J. Biomed. Health Inform.* 26 (8), 4044–4055.
- Tan, W., Zhou, L., Li, X., Yang, X., Chen, Y., Yang, J., 2021. Automated vessel segmentation in lung CT and CTA images via deep neural networks. *J. Xray Sci. Technol.* 29 (6), 1123–1137.
- Valanarasu, J. M.J., Oza, P., Hachililoglu, I., Patel, V.M., 2021. Medical transformer: gated axial-attention for medical image segmentation. In: *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, pp. 36–46.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I., 2017. Attention is all you need. *Adv. Neural. Inf. Process. Syst.* 30.
- Virani, S.S., Alonso, A., Aparicio, H.J., Benjamin, E.J., Bittencourt, M.S., Callaway, C.W., Carson, A.P., Chamberlain, A.M., Cheng, S., Delling, F.N., et al., 2021. Heart disease and stroke statistics–2021 update: a report from the American Heart Association. *Circul.* 143 (8), e254–e743.
- Wang, H., Guo, S., Ye, J., Deng, Z., Cheng, J., Li, T., Chen, J., Su, Y., Huang, Z., Shen, Y., et al., 2023. Sam-med3d: towards general-purpose segmentation models for volumetric medical images. *arXiv prepr.*
- Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., Hu, Q., 2020. Eca-net: efficient channel attention for deep convolutional neural networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11534–11542.
- Wang, S., Li, B., Zhou, S., 2012. A segmentation method of coronary angiograms based on multi-scale filtering and region-growing. In: *2012 International Conference on Biomedical Engineering and Biotechnology*. IEEE, pp. 678–681.
- Wu, H., Chen, S., Chen, G., Wang, W., Lei, B., Wen, Z., 2022. Fat-net: feature adaptive transformers for automated skin lesion segmentation. *Med Image Anal.* 76, 102327.
- Xu, J., Dong, A., Yang, Y., Jin, S., Zeng, J., Xu, Z., Jiang, W., Zhang, L., Dong, J., Wang, B., 2025. Vsnets: vessel structure-aware network for hepatic and portal vein segmentation. *Med Image Anal.* 101, 103458.
- Xu, Y., Tang, J., Men, A., Chen, Q., 2024. Eviprompt: a training-free evidential prompt generation method for adapting segment anything model in medical images. *IEEE Trans. Image Process.*
- Yang, G., Kitslaar, P., Frenay, M., Broersen, A., Boegers, M.J., Bax, J.J., Reiber, J. H.C., Dijkstra, J., 2012. Automatic centerline extraction of coronary arteries in coronary computed tomographic angiography. *Int. J. Cardiovasc. Imaging* 28, 921–933.
- Yang, X., Xu, L., Yu, S., Xia, Q., Li, H., Zhang, S., 2024. Segmentation and vascular vectorization for coronary artery by geometry-based cascaded neural network. *IEEE Trans. Med. Imaging*.
- Ye, H., Zhang, X., Hu, Y., Fu, H., Liu, J., 2025. Vsr-net: vessel-like structure rehabilitation network with graph clustering. *IEEE Trans. Image Process.*
- Yu, A., Gao, K., You, X., Zhong, Y., Su, Y., Liu, B., Qiu, C., 2025. Rethinking semantic segmentation with multi-grained logical prototype. *IEEE Trans. Image Process.*
- Zeng, A., Wu, C., Lin, G., Xie, W., Hong, J., Huang, M., Zhuang, J., Bi, S., Pan, D., Ullah, N., et al., 2023. ImageCAS: a large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images. *Comput. Med. Imaging Graph.* 109, 102287.
- Zhang, C., Cai, Y., Lin, G., Shen, C., 2020. DeepEMD: few-shot image classification with differentiable earth mover's distance and structured classifiers. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 12203–12213.
- Zhang, M., Wu, Y., Zhang, H., Qin, Y., Zheng, H., Tang, W., Arnold, C., Pei, C., Yu, P., Nan, Y., et al., 2023. Multi-site, multi-domain airway tree modeling. *Med Image Anal.* 90, 102957.
- Zhao, G., Liang, K., Pan, C., Zhang, F., Wu, X., Hu, X., Yu, Y., 2022. Graph convolution based cross-network multiscale feature fusion for deep vessel segmentation. *IEEE Trans. Med. Imaging* 42 (1), 183–195.
- Zhou, H.-Y., Guo, J., Zhang, Y., Han, X., Yu, L., Wang, L., Yu, Y., 2023. nnformer: volumetric medical image segmentation via a 3d transformer. *IEEE Trans. Image Process.*
- Zhou, L., Jiang, Y., Li, W., Hu, J., Zheng, S., 2024a. Shape-scale co-awareness network for 3d brain tumor segmentation. *IEEE Trans. Med. Imaging*.
- Zhou, L., Zhang, Y., Zhang, J., Qian, X., Gong, C., Sun, K., Ding, Z., Wang, X., Li, Z., Liu, Z., et al., 2024b. Prototype learning guided hybrid network for breast tumor segmentation in DCE-MRI. *IEEE Trans. Med. Imaging*.